

Network Systems Design

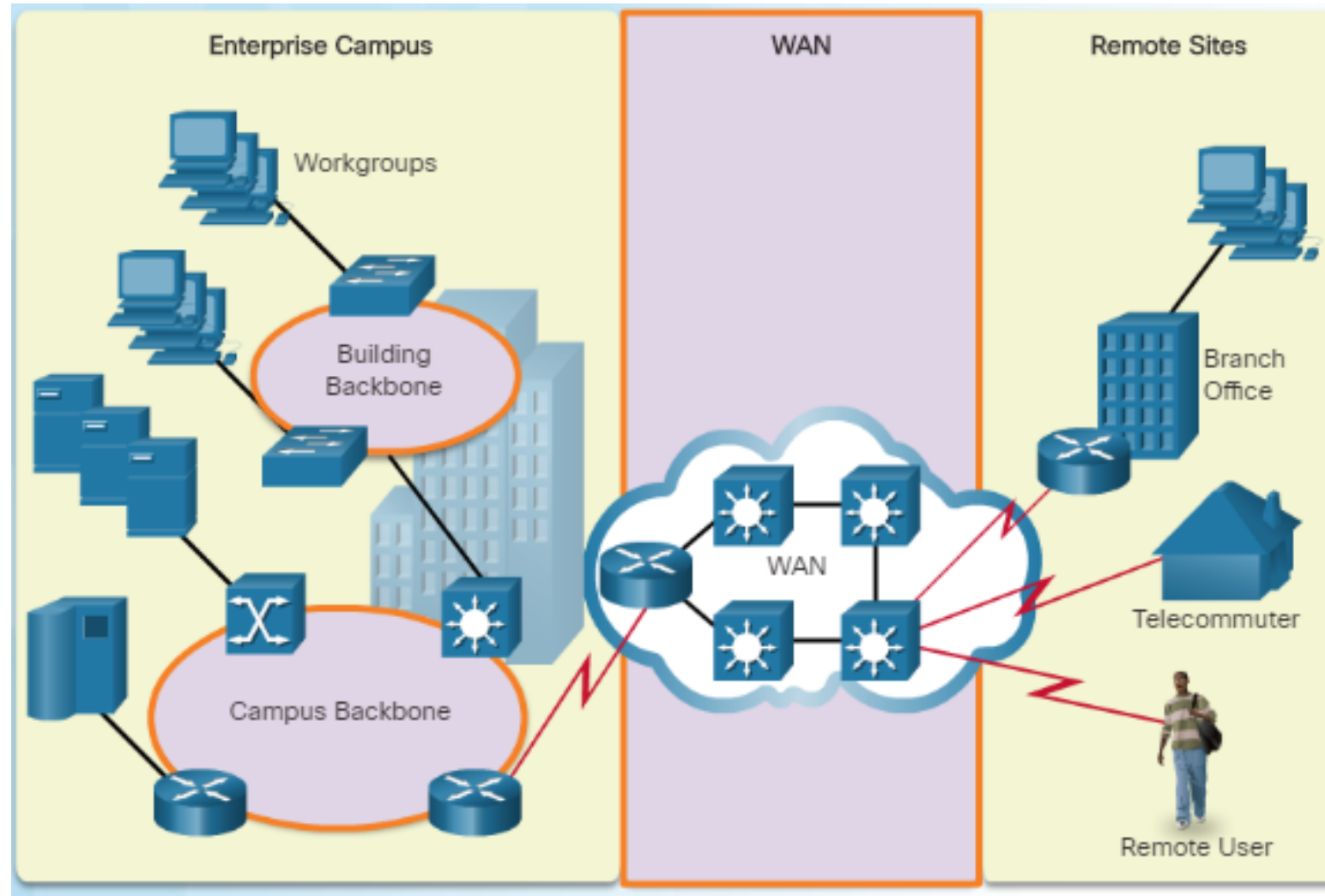
Year II

Wide Area Networks

Outline

- WAN
- WAN Topologies
- WAN in ISO/OSI
- WAN Terminologies
- WAN Devices
- WAN Link Connections
- How to select a WAN link connection

WIDE AREA NETWORK (WAN)



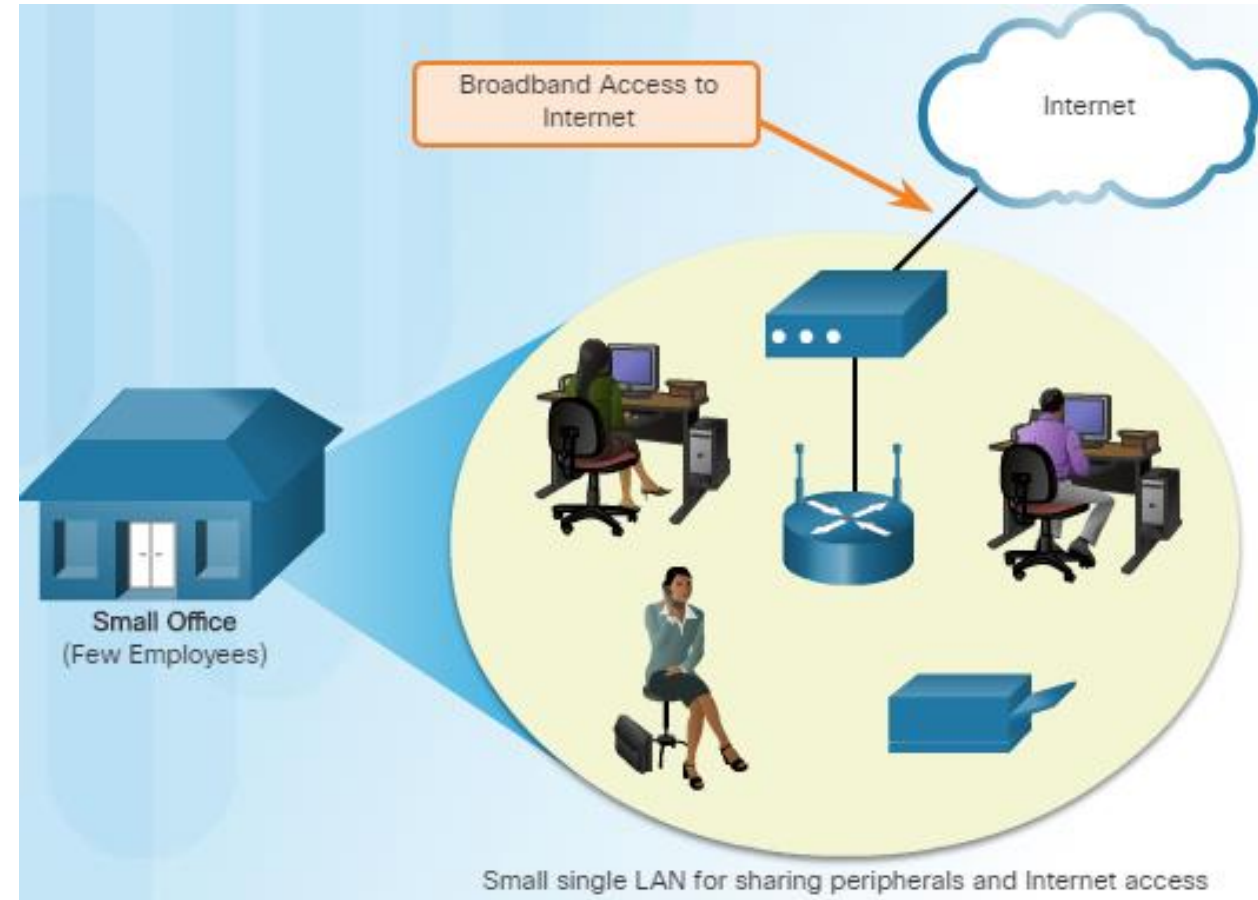
WIDE AREA NETWORK (WAN) CONT.

- Operates beyond the geographic scope of a LAN.
- Used to interconnect the enterprise LAN to remote LANs in branch sites.
- Owned by a service provider (LAN is owned by an organization).
- Service providers provide links to interconnect remote sites for the purpose of transporting data, voice, and video
- An organization must pay a fee to use the WAN service provider's network services to connect remote sites.

WIDE AREA NETWORK (WAN) CONT.

Small Office

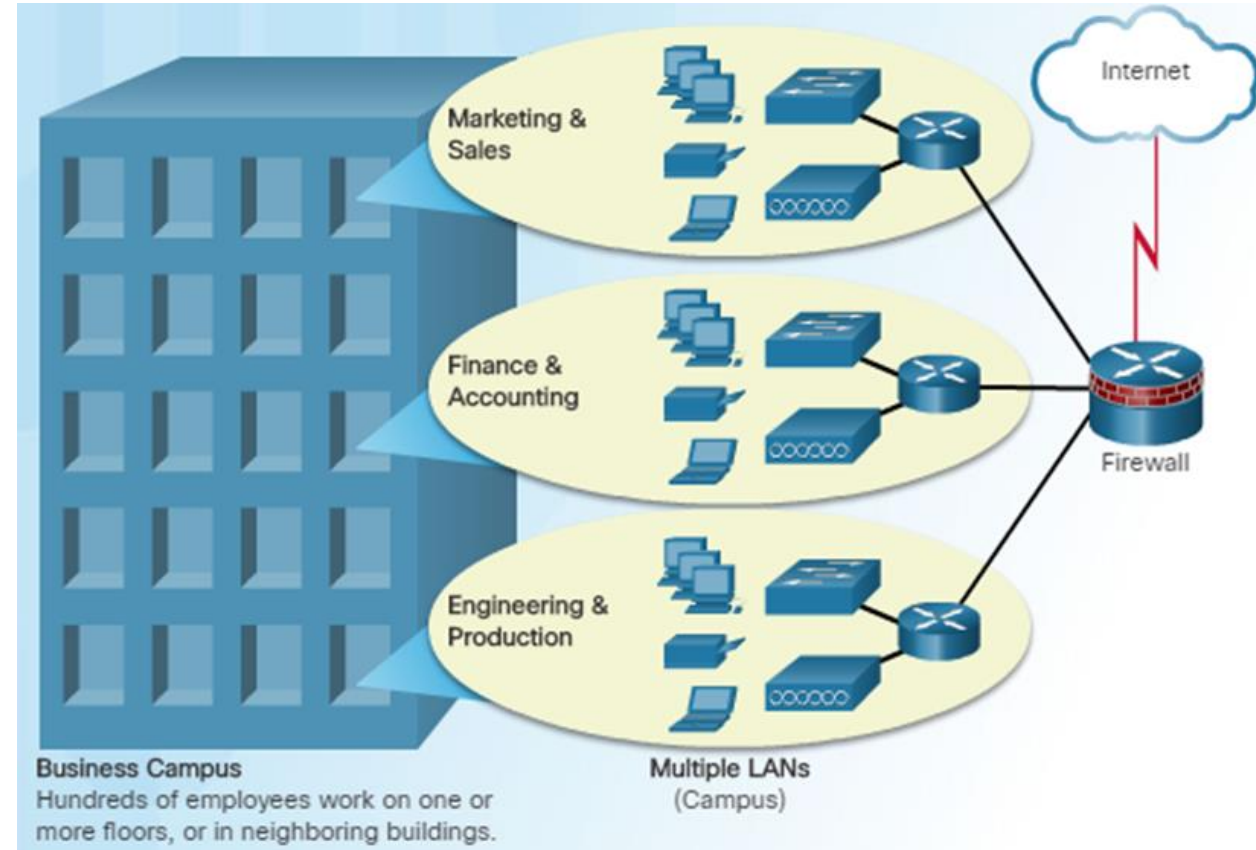
- Connects to the Internet using DSL



WIDE AREA NETWORK (WAN) CONT.

Campus Network

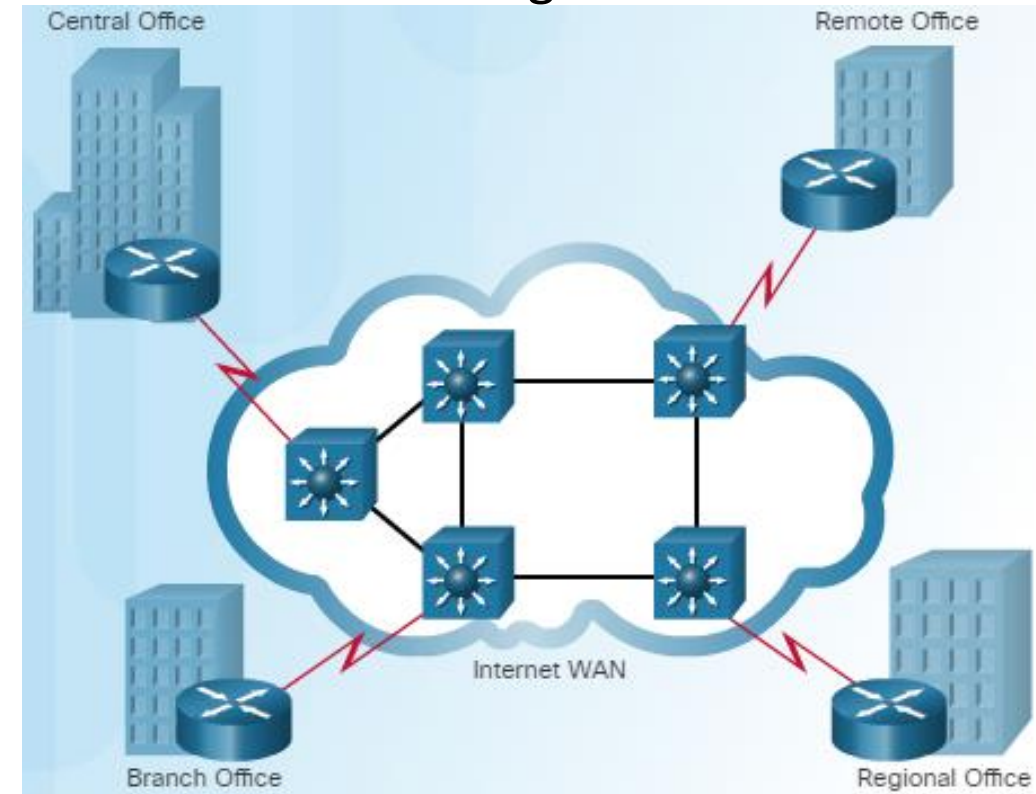
- Connects to the Internet using DSL



WIDE AREA NETWORK (WAN) CONT.

Branch Network

- Required a WAN for the remote sites to access the data center
- The branch offices that are in nearby cities use private dedicated lines through their local service provider.
- Offices that are located in other countries must use the Internet for their WAN connection.



WIDE AREA NETWORK (WAN) CONT.

Distributed Network

- Site-to-site and remote access Virtual Private Networks (VPNs) enable the company to use the Internet to connect easily and securely



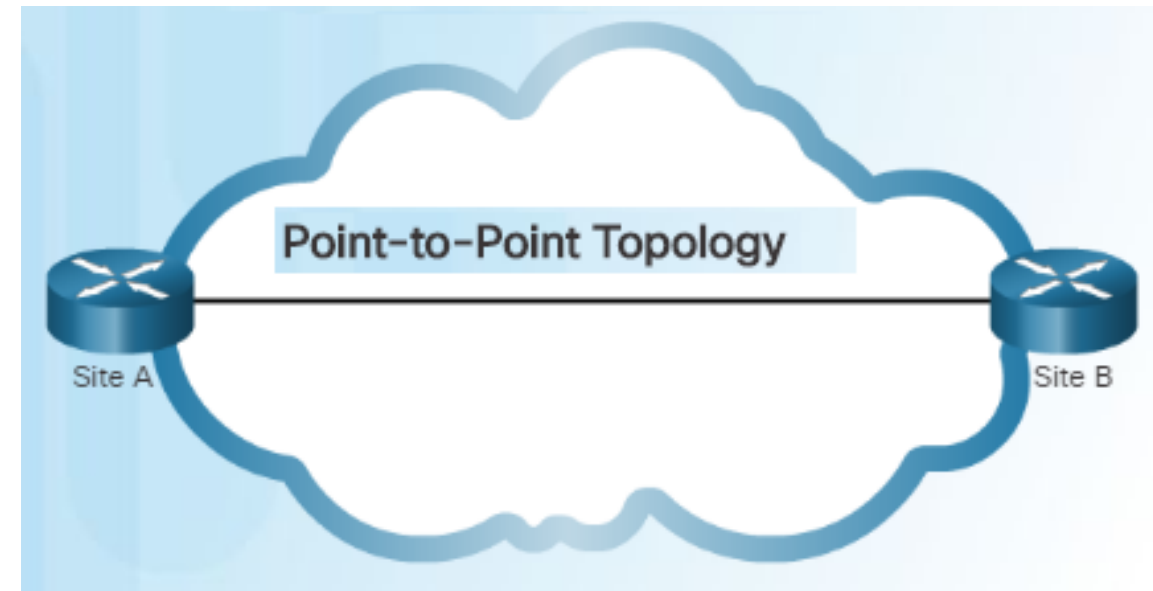
WAN TOPOLOGIES

- Point-to-Point topology
- Hub-and-Spoke
- Full meshed
- Dual-Homed

WAN TOPOLOGIES CONT.

Point-to-Point topology

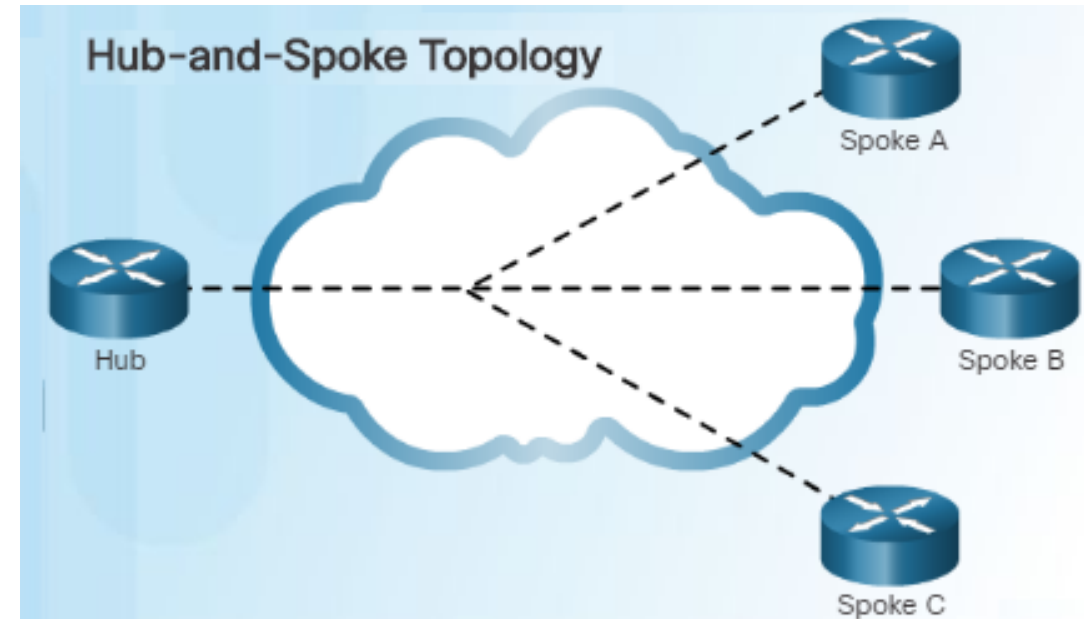
- Employs a point-to-point circuit between two endpoints
- Involves a dedicated leased-line connection such as a T1/E1 line.
- Transparent to the customer network and appears to be a direct physical link between two endpoints



WAN TOPOLOGIES CONT.

Hub-and-Spoke

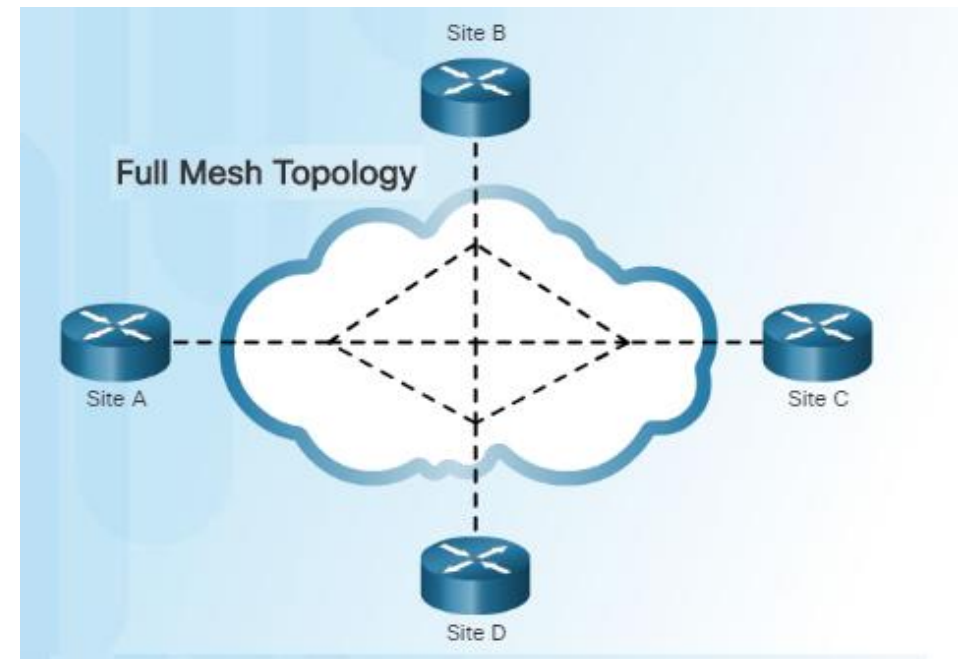
- Applicable when a private network connection between multiple sites is required
- A single interface to the hub can be shared by all spoke circuits (point-to-multipoint)
- Spoke sites can be interconnected through the hub site using virtual circuits and routed sub-interfaces at the hub.
- Disadvantage:
All communication has to go through the hub.



WAN TOPOLOGIES CONT.

Full Mesh

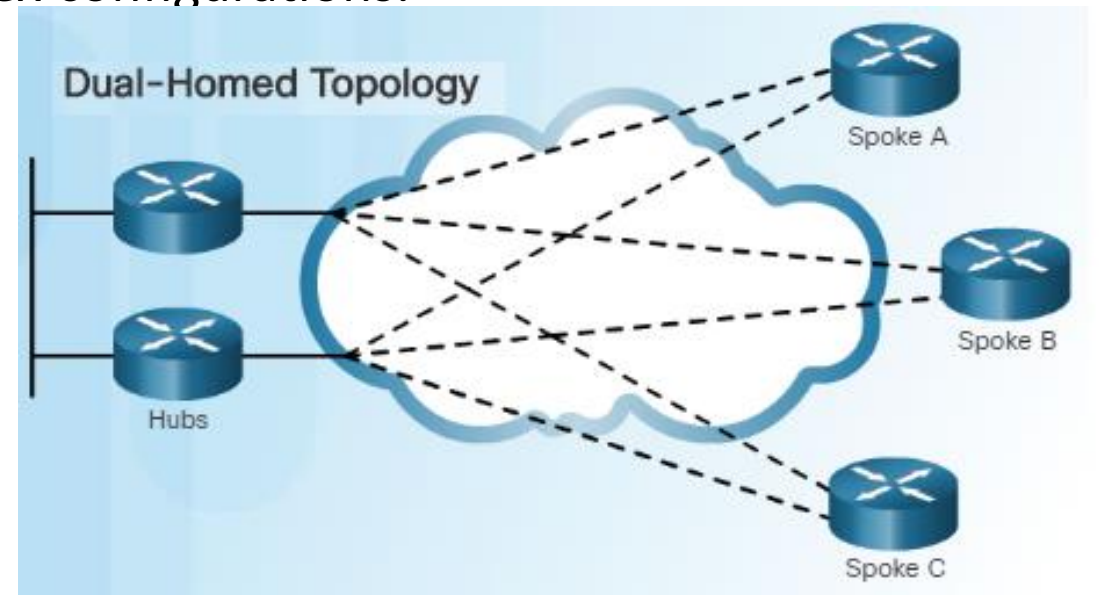
- With a full mesh topology using virtual circuits, any site can communicate directly with any other site.
- Disadvantage
 - Large number of virtual circuits need to be configured and maintained.



WAN TOPOLOGIES CONT.

Dual-homed Topology

- Provides redundancy and load balancing
- More expensive to implement
- Requires additional networking hardware including routers and switches.
- More difficult to implement since they require complex configurations.



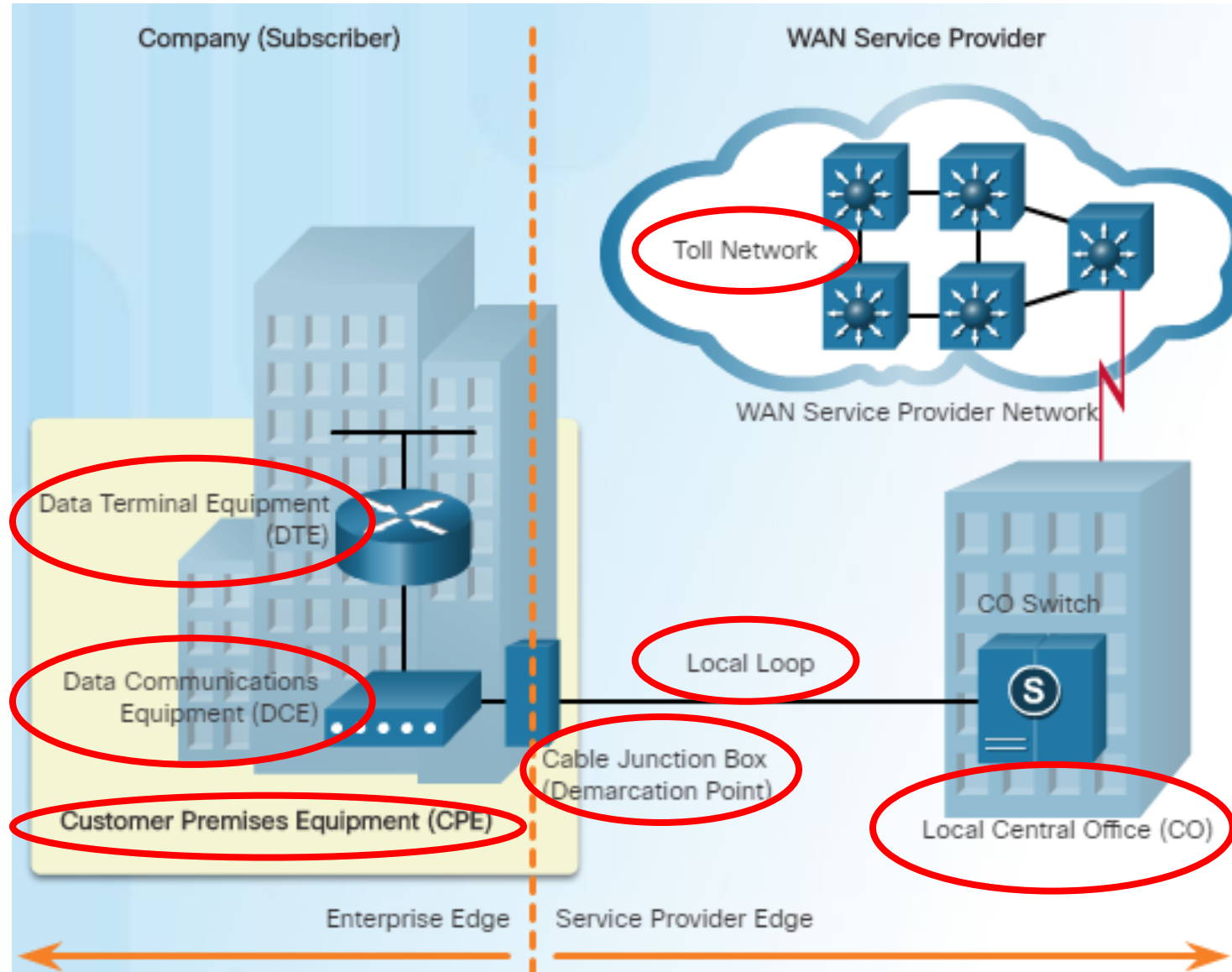
WANS IN THE ISO/OSI MODEL

- Operates on the physical and data link layer of the OSI Model.
- Physical layer protocols describe how to provide electrical, mechanical, operational, and functional connects to the services of a communications service provider.
- WAN access standards are defined and managed by a number of recognized authorities:
 - TIA/EIA (Telecommunications Industry Association and the Electronic Industries Alliance)
 - ISO (International Organization for Standardization)
 - IEEE (Institute of Electrical and Electronics Engineers)

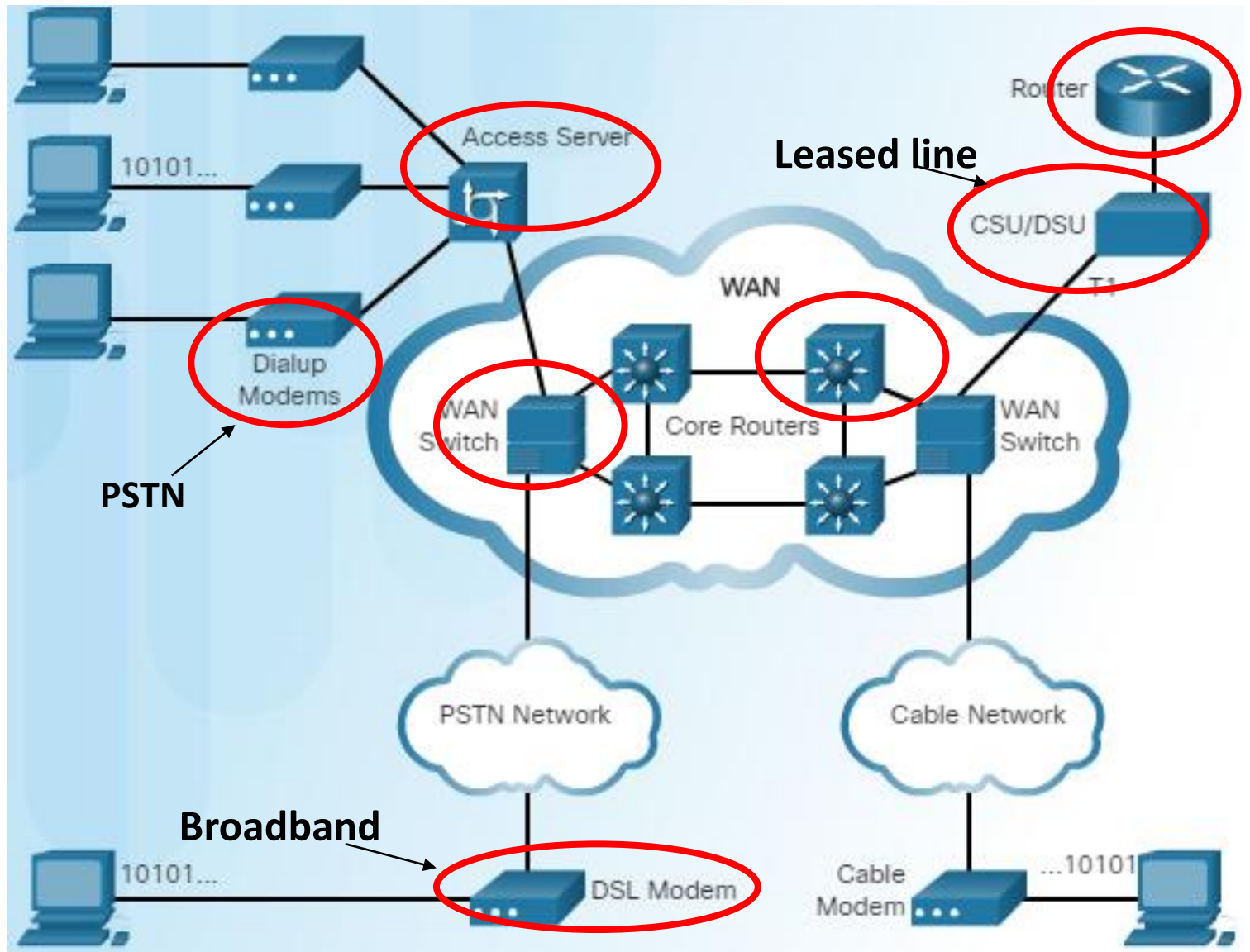
WANS IN THE ISO/OSI MODEL CONT.

- Data link layer protocols define how data is encapsulated and the mechanisms for transferring the resulting frames.
- Example protocols are:
HDLC, PPP, Frame relay, Ethernet WANs, MPLS, VSAT, Broadband
- Data link layer requirements include physical addressing, flow control and encapsulation.

WAN TERMINOLOGY



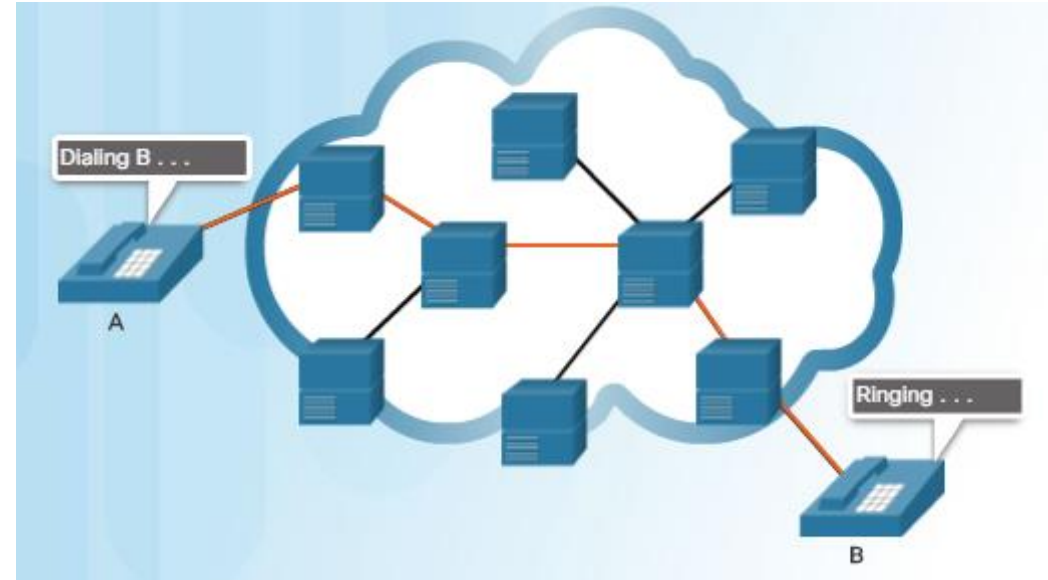
WAN DEVICES



WAN OPERATIONS

Circuit Switching

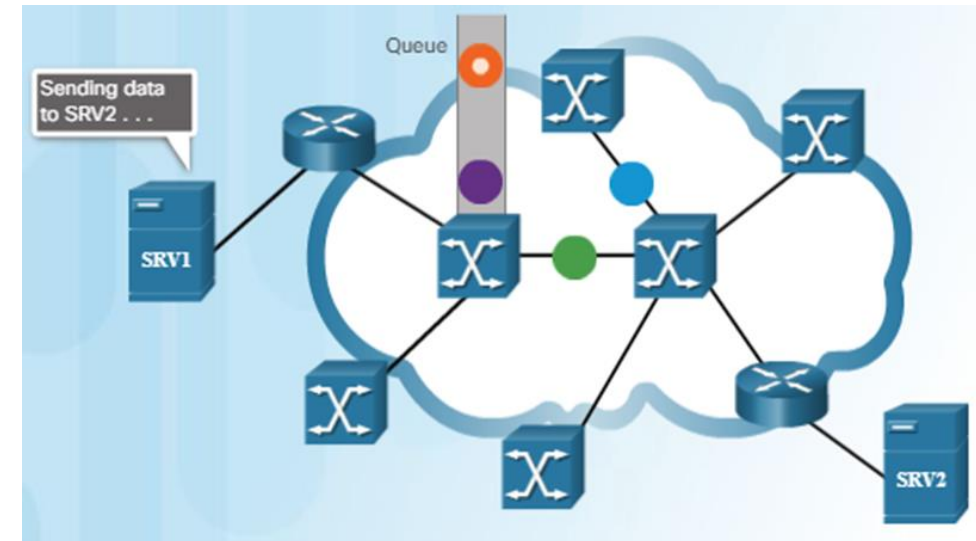
- Establishes a dedicated circuit (or channel) (virtual) between nodes before the data transfer.
- A dedicated path (set of links) is allocated for each data transfer.
- Circuit Establishment → Data transfer → Circuit disconnect.
- Data is processed at source to decide the entire path till destination.
- Ex: Dialing a number to make a call
 - ✓ Public Switched Telephone Network (PSTN)
 - ✓ Integrated Services Digital Network (ISDN)



WAN OPERATIONS CONT.

Packet Switching

- No dedicated circuit (or channel) (virtual) established between nodes before the data transfer.
- The intermediate path is decided by the routers on the go.
- Packets belong to same communication may follow different paths.
- Less wastage of resources, but less reliable
- Known as **Connectionless systems (datagram switching)**
version of Packet Switching

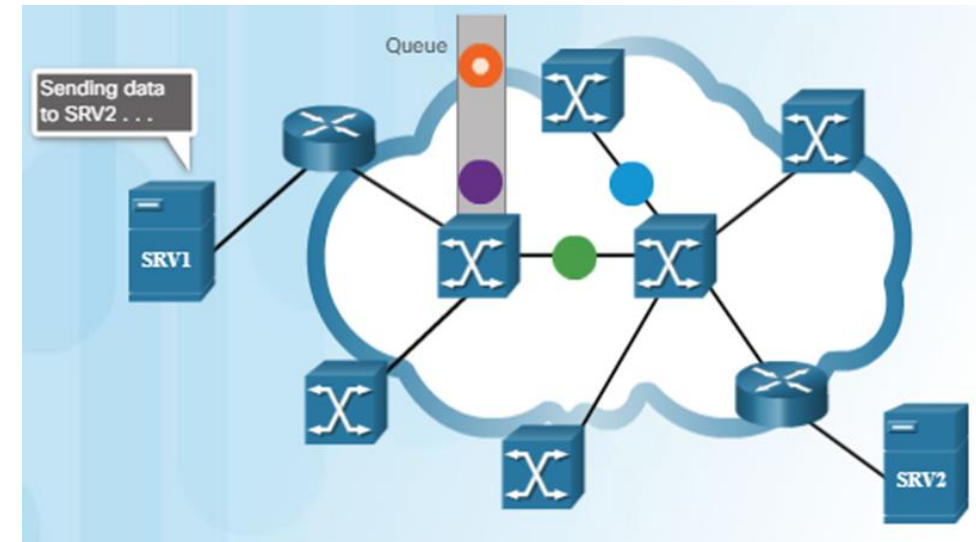


WAN OPERATIONS CONT.

Packet Switching

Connectionless systems (datagram switching)

- Full addressing information (destination address) must be carried in each packet.
- The packets are routed at each router individually, based on addressing information.
- Ex: Internet



WAN OPERATIONS CONT.

Packet Switching Cont.

- Circuit switching and Connectionless systems of Packet Switching are two extremes:
 - with a dedicated path for each data transfer and without even a path selected
- It is worth to have a technique which is in between the two extremes.
- Let a path (set of links) from source to destination be allocated for a data transfer,
 - but the links should not be dedicated only to that data transfer
- The path (set of links) can be shared by other data transfers.
- Known as **Connection-oriented systems (virtual circuit switching)** version of Packet Switching

WAN OPERATIONS CONT.

Packet Switching Cont.

Connection-oriented systems (virtual circuit switching)

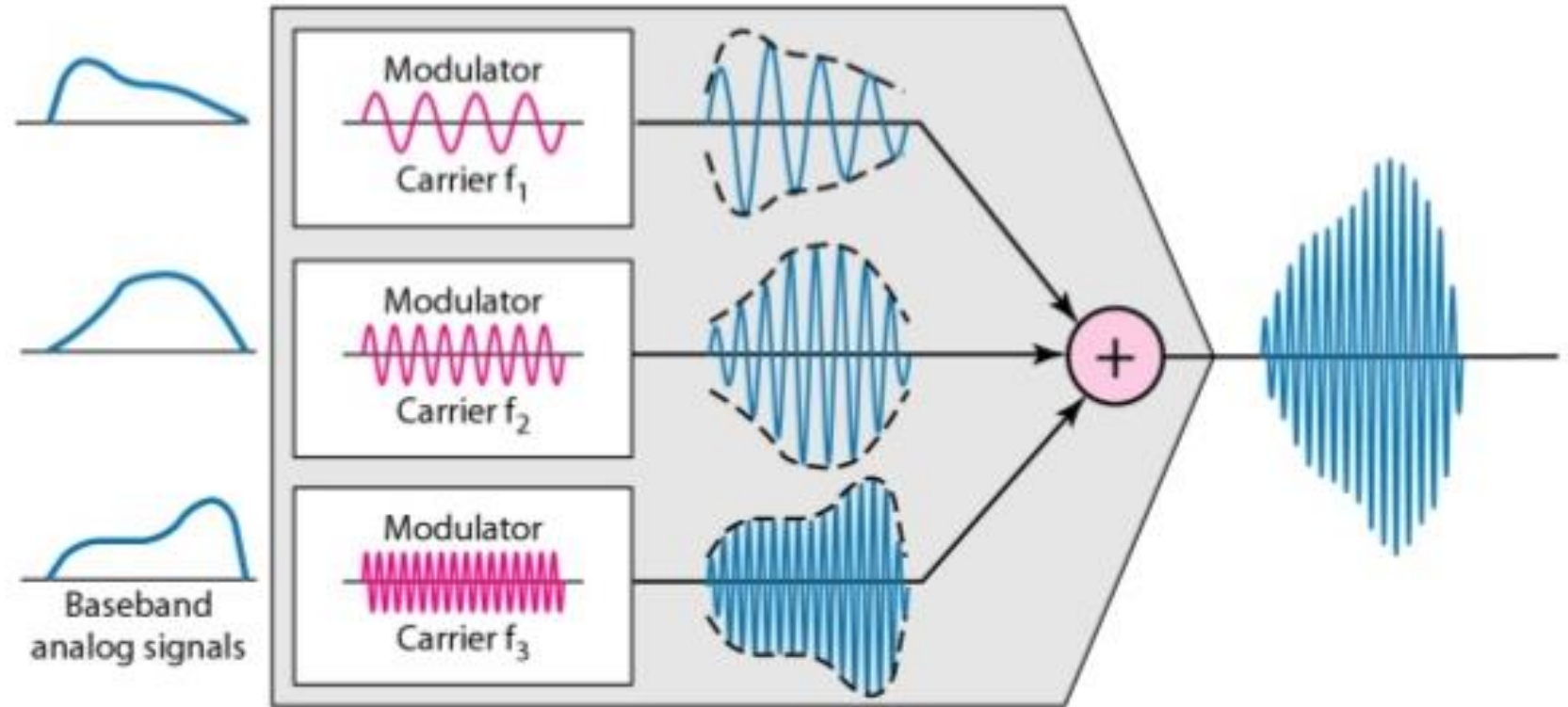
- ✓ All packets of a single data transfer will travel through the allocated path, when the links of that path are free.
- ✓ The packets have to wait at each node until the forward link is free and available.
- ✓ Every packet does not need to carry addressing information, as the path is pre-defined.
- ✓ Every path carries path numbers as an identifies for the path.
- ✓ Ex: Frame Relay

MULTIPLEXING

- Sending multiple signals of information over a single communications link at the same time in the form of a single, complex signal.
 - ✓ Frequency Division Multiplexing (FDM)
 - ✓ Time Division Multiplexing (TDM)
 - ✓ Wave length Division Multiplexing (WDM)

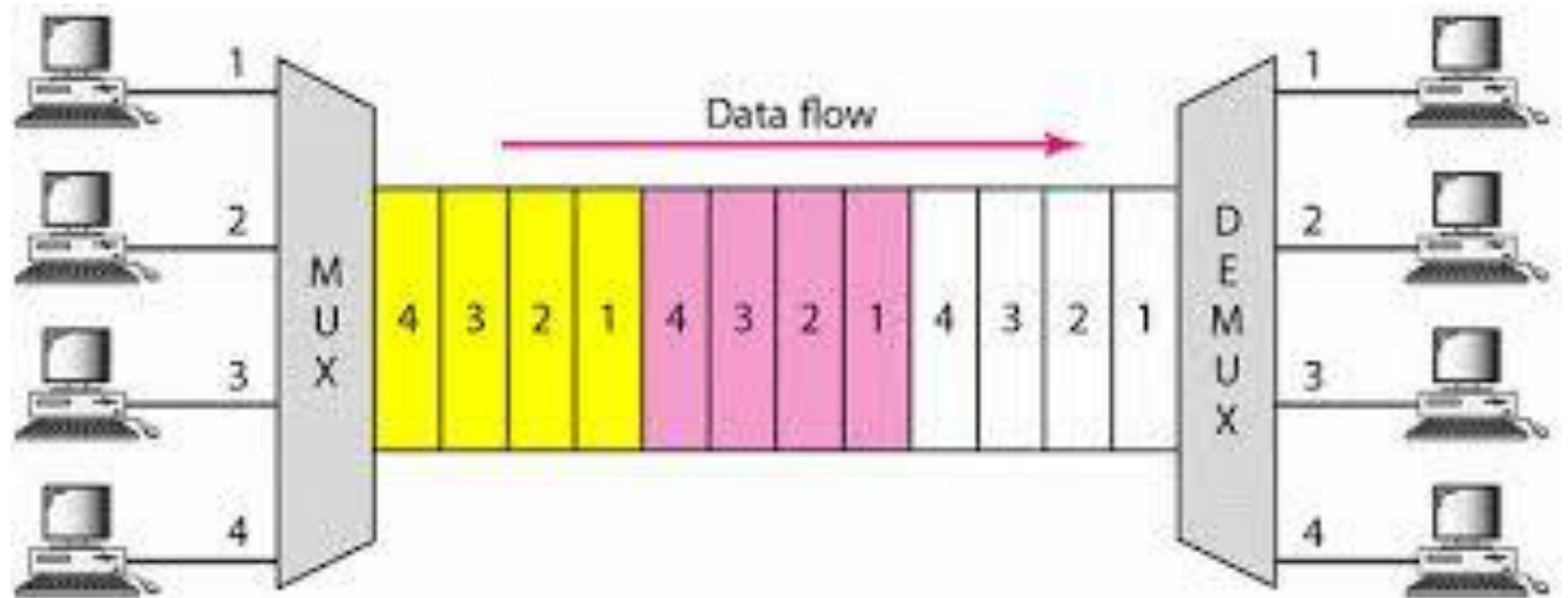
multiplexing

Frequency Division Multiplexing (FDM)

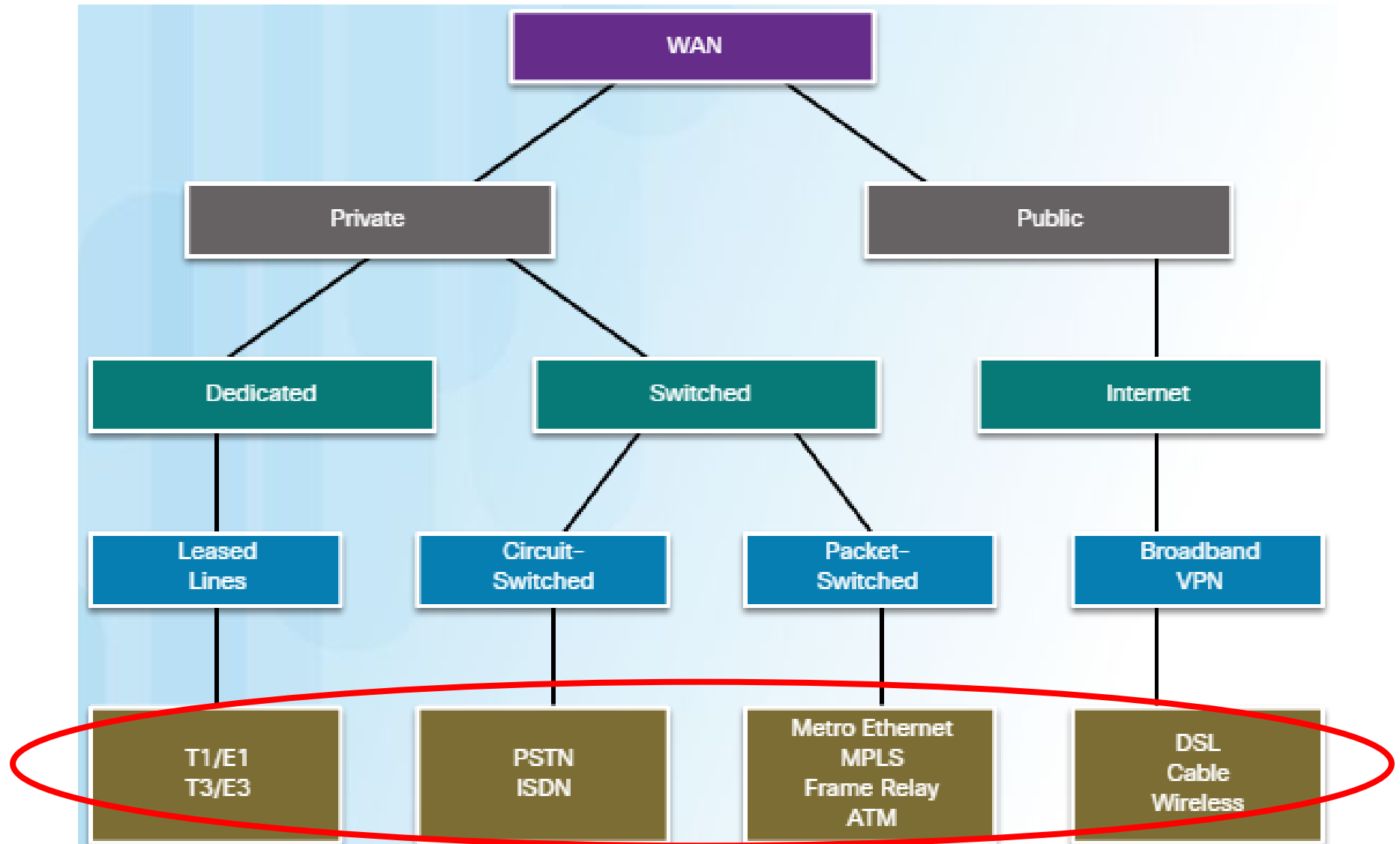


multiplexing

Time Division Multiplexing (TDM)

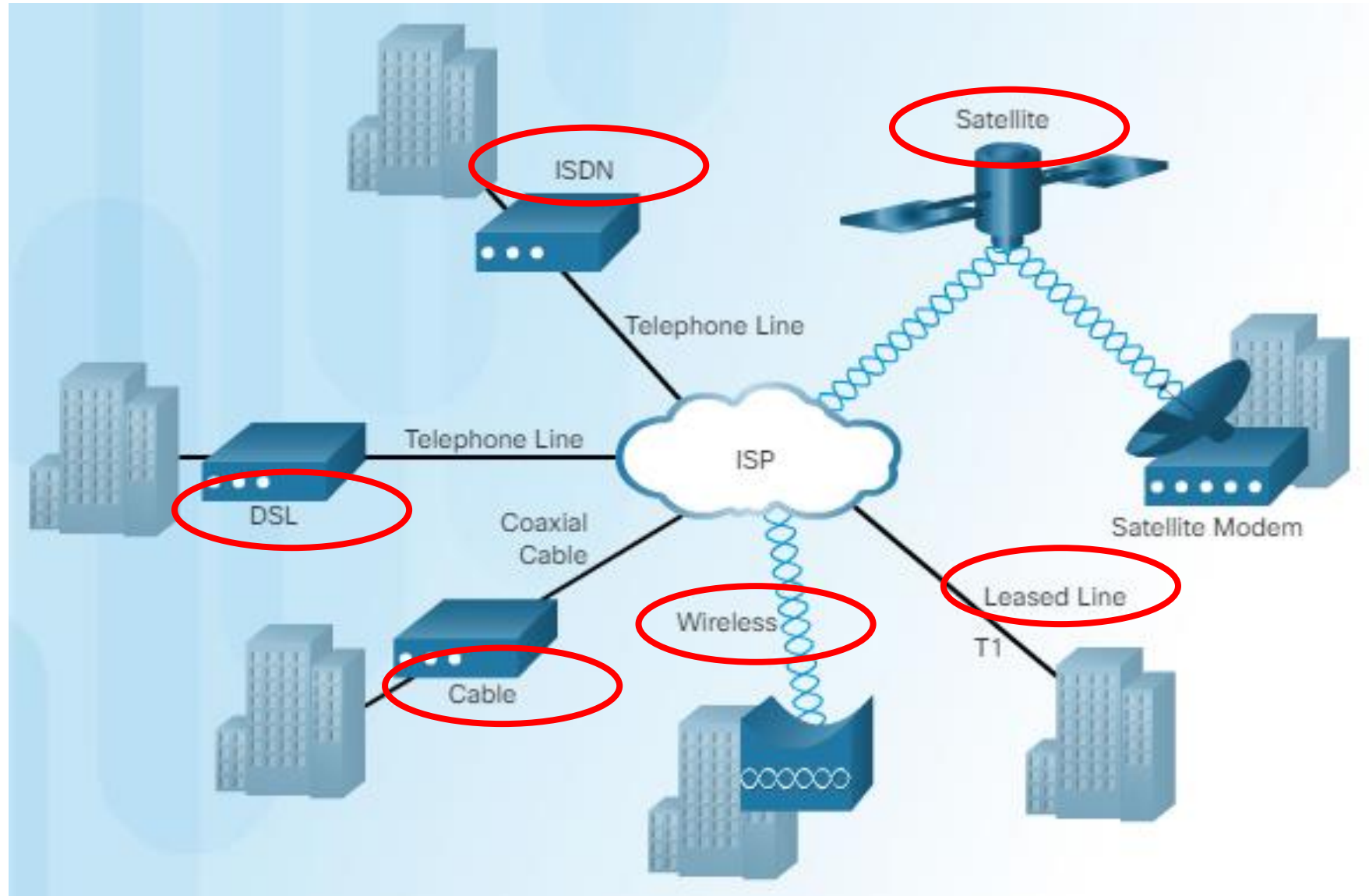


Wan link connections options



**WAN Access
Technologies**

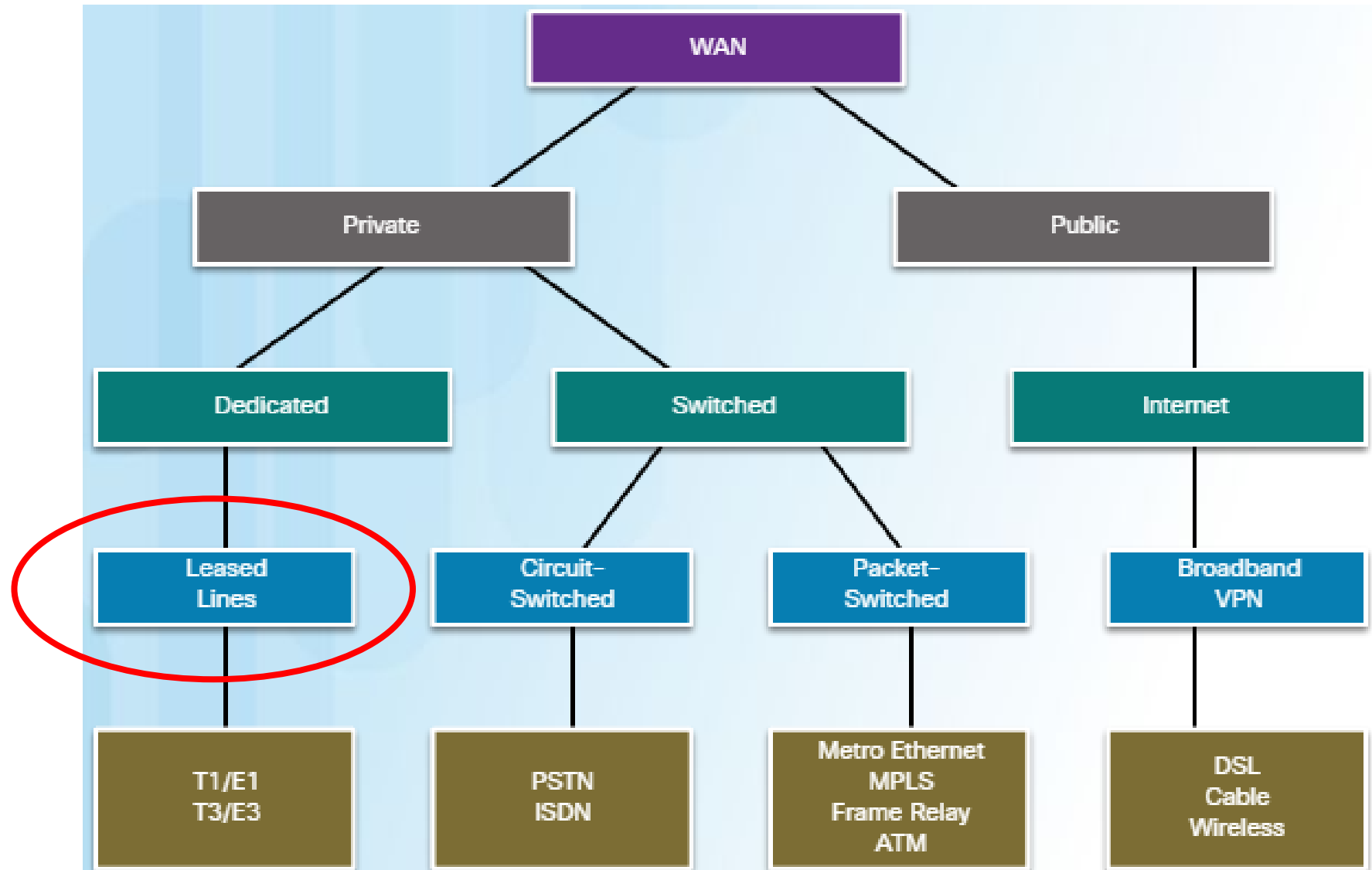
Wan access technologies



Service Provider Network Infrastructure

- Complex and consist of high-bandwidth fiber-optic media, using Synchronous Optical Networking (SONET) or Synchronous Digital Hierarchy (SDH) standard.
- A newer fiber-optic media development for long-range communications is called dense wavelength division multiplexing (DWDM).

Private WAN Infrastructures

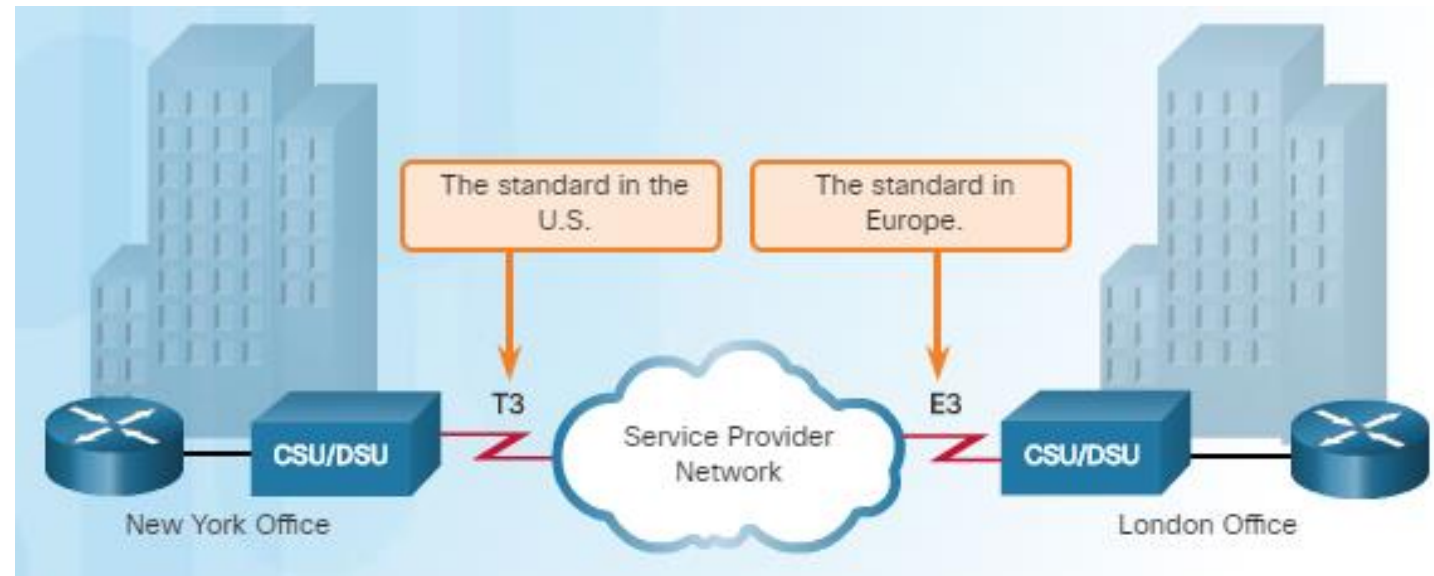


Private WAN Infrastructures cont.

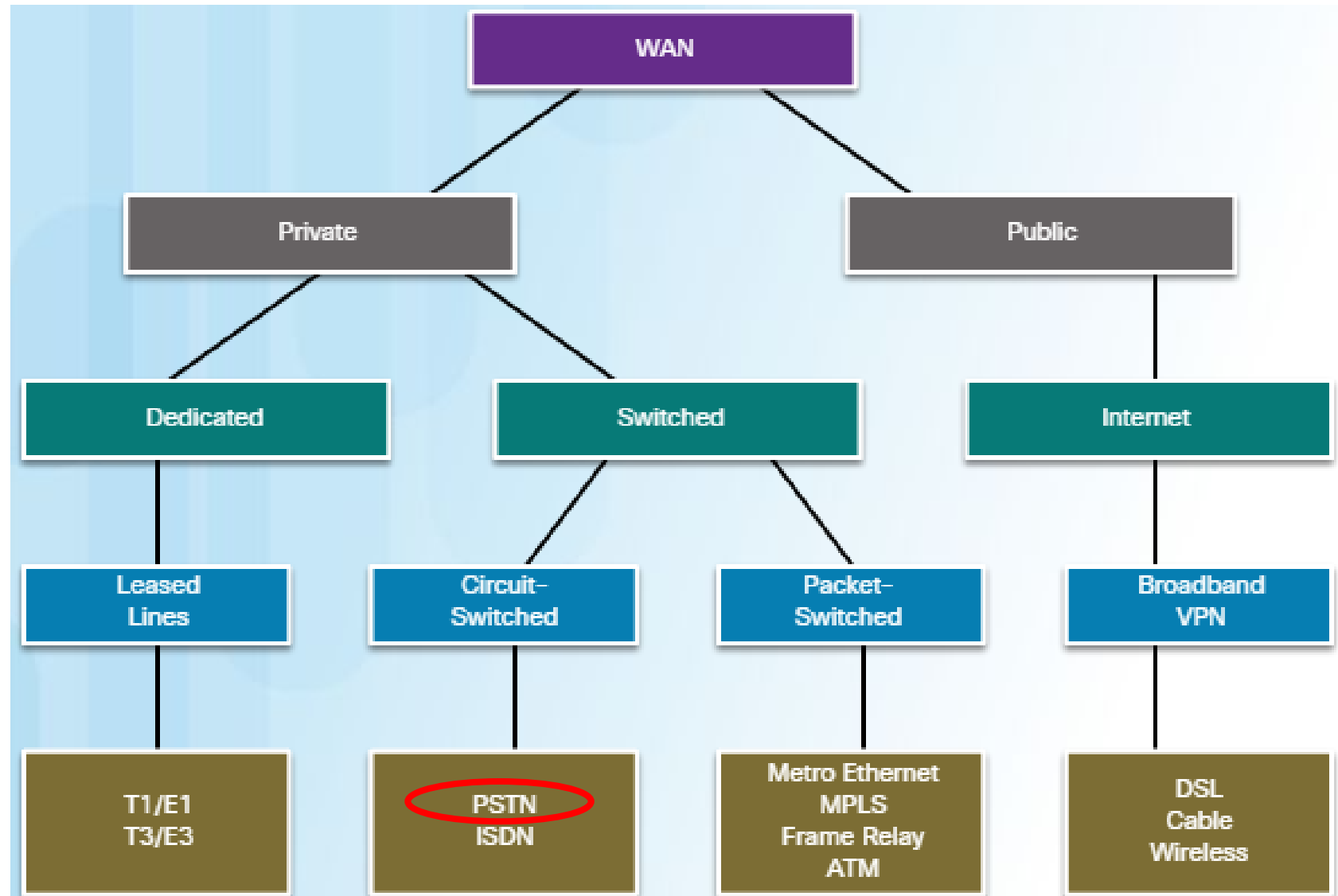
Private, Dedicated

Leased Lines

- When permanent dedicated connections are required, a point-to-point link is used to provide a pre-established WAN communications
- Always active
- Price depends on the bandwidth required and the distance between the two connected paths.



Private WAN Infrastructures cont.

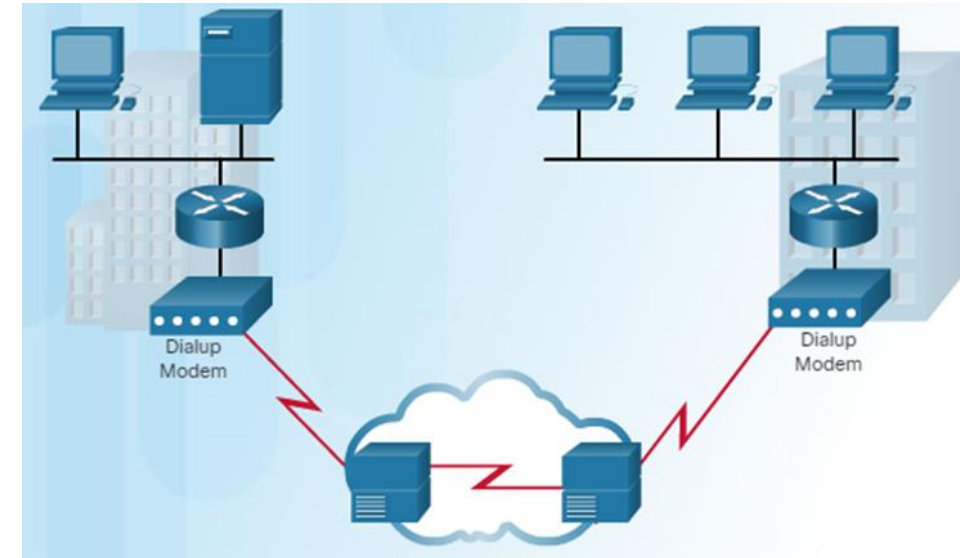


Private WAN Infrastructures cont.

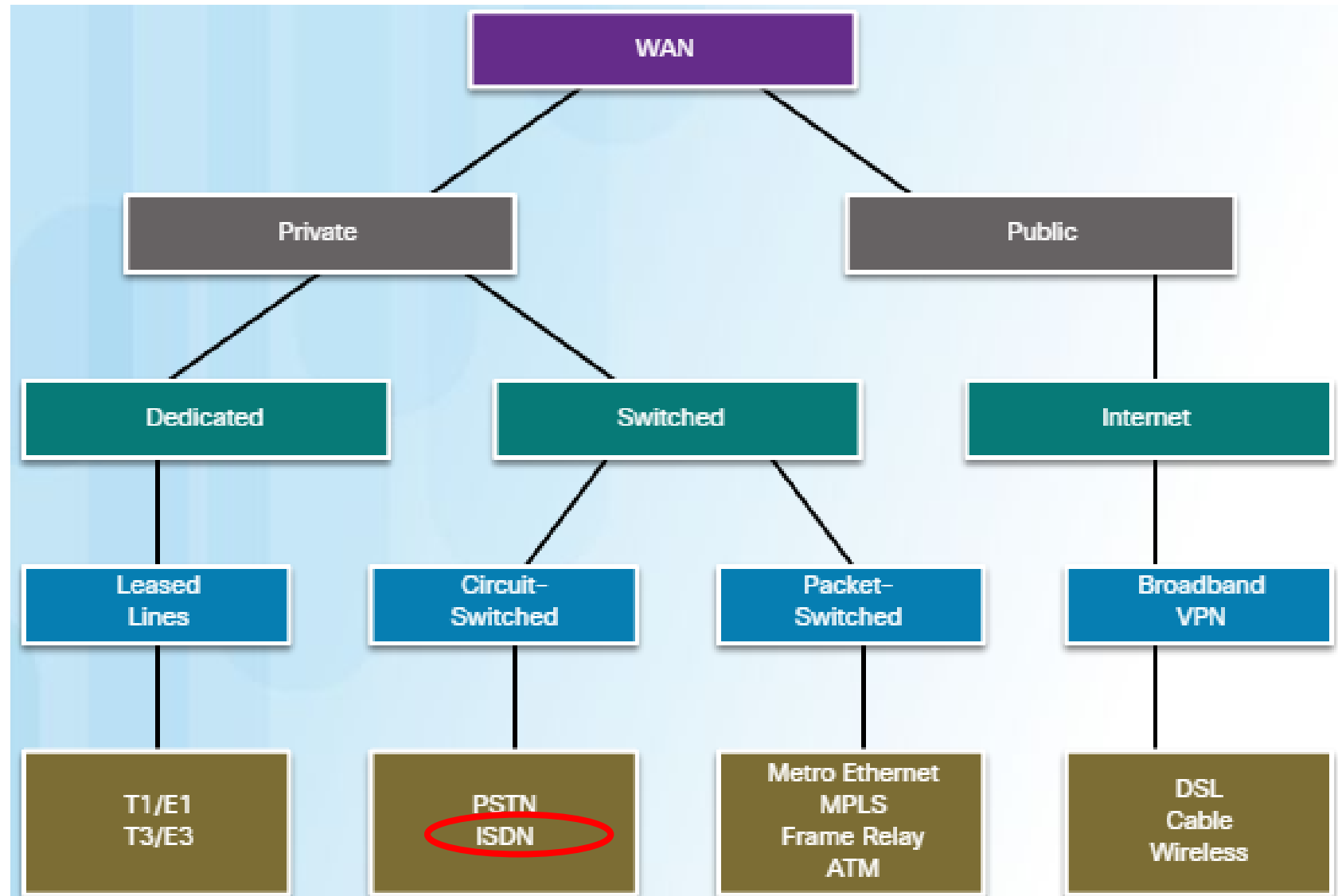
Private, Switched,
Circuit - switched

PSTN (Dial up)

- Analog transmission of data over the existing PSTN as a dial up service.
- WAN built with an on demand connection (dial up)
 - using a modem and the existing voice telephone network (PSTN)
- A modem modulates binary data into an analog signal at the source and demodulates at the destination.
- Use one line for one conversation at a time,
 - using one phone number.
- Cannot be used for any other purpose
 - while a call is being made.



Private WAN Infrastructures cont.

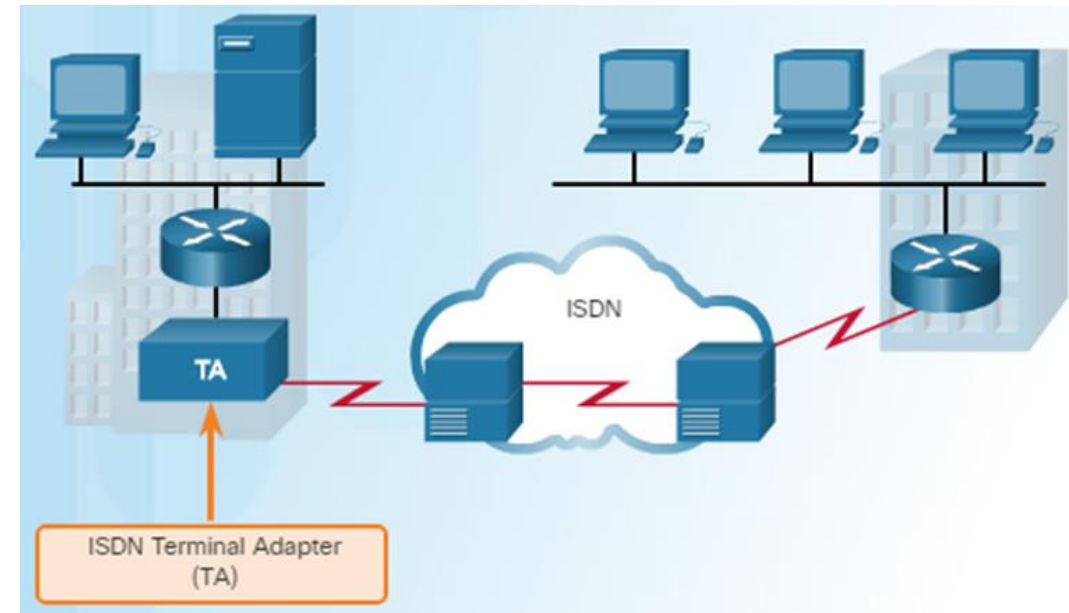


Private WAN Infrastructures cont.

Private, Switched,
Circuit - switched

Integrated Services Digital Network (ISDN)

- Digital transmission of data over the existing PSTN as a dial up service.
- ISDN changes the internal connections of the PSTN from carrying analog signals to time-division multiplexed (TDM) digital signals.
- Can be used as a backup if the leased line fails
- Defines access interfaces with **dedicated bandwidth**:
 - ✓ Basic Rate Interface (BRI)
 - ✓ Primary Rate Interface (PRI)

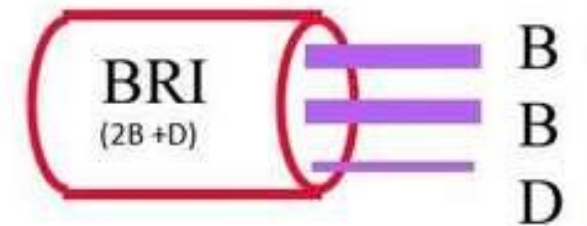


Private WAN Infrastructures cont.

Integrated Services Digital Network (ISDN) Cont.

ISDN BRI

- For home users or smaller organizations
- Provides two 64 kb/s bearer channels (B) for carrying voice and data and a 16 kb/s delta channel (D) for signaling, call setup and other purposes.

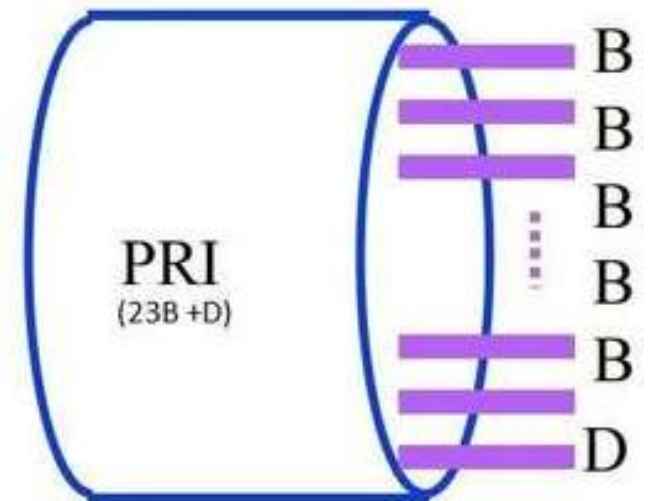


Private WAN Infrastructures cont.

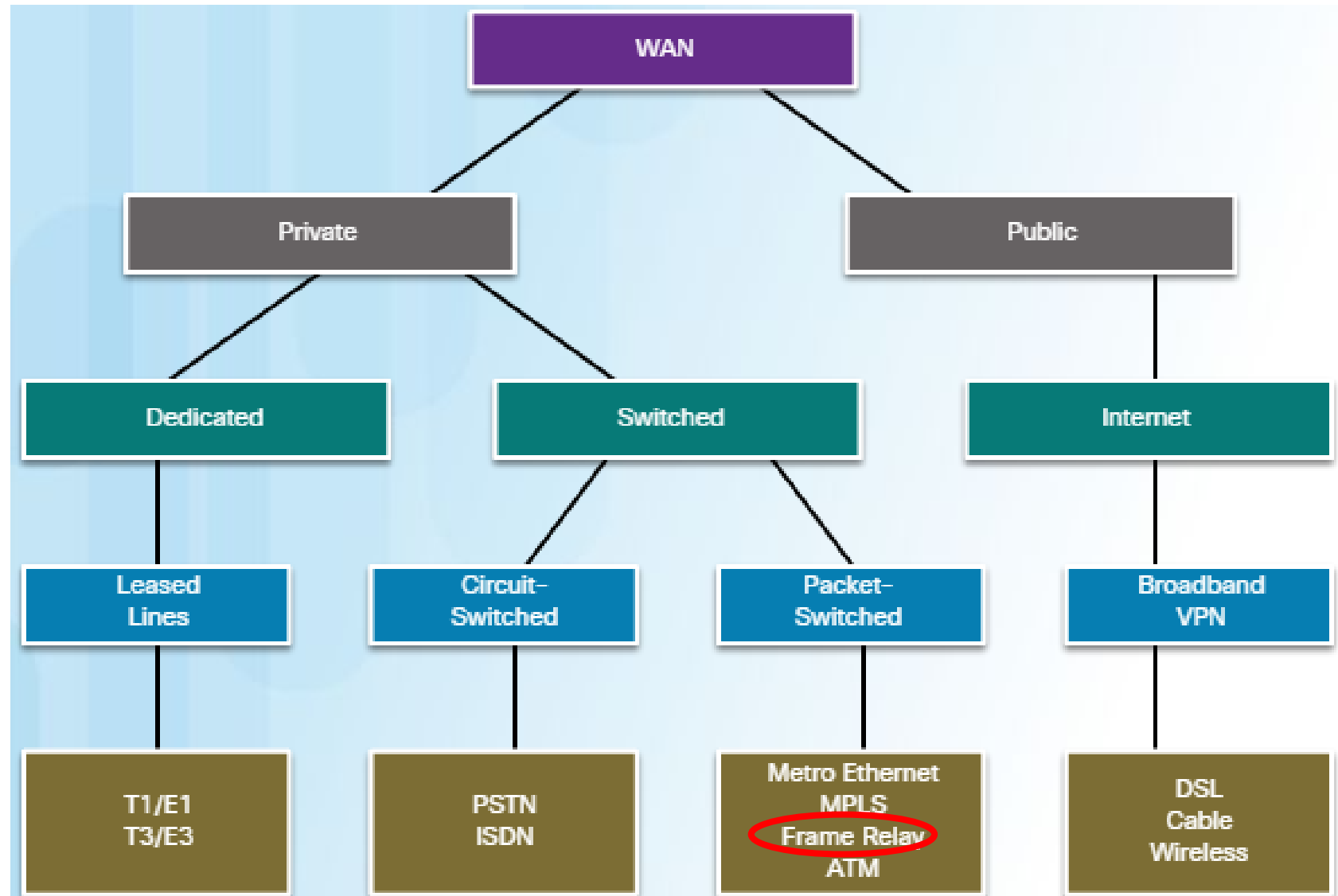
Integrated Services Digital Network (ISDN) Cont.

ISDN PRI

- Used by medium to large sized organizations
- Provides 23 64 kb/s bearer channels (B) for carrying voice and data
and a 64 kb/s delta channel (D) for signaling, call setup and other purposes



Private WAN Infrastructures cont.



Private WAN Infrastructures cont.

Frame Relay

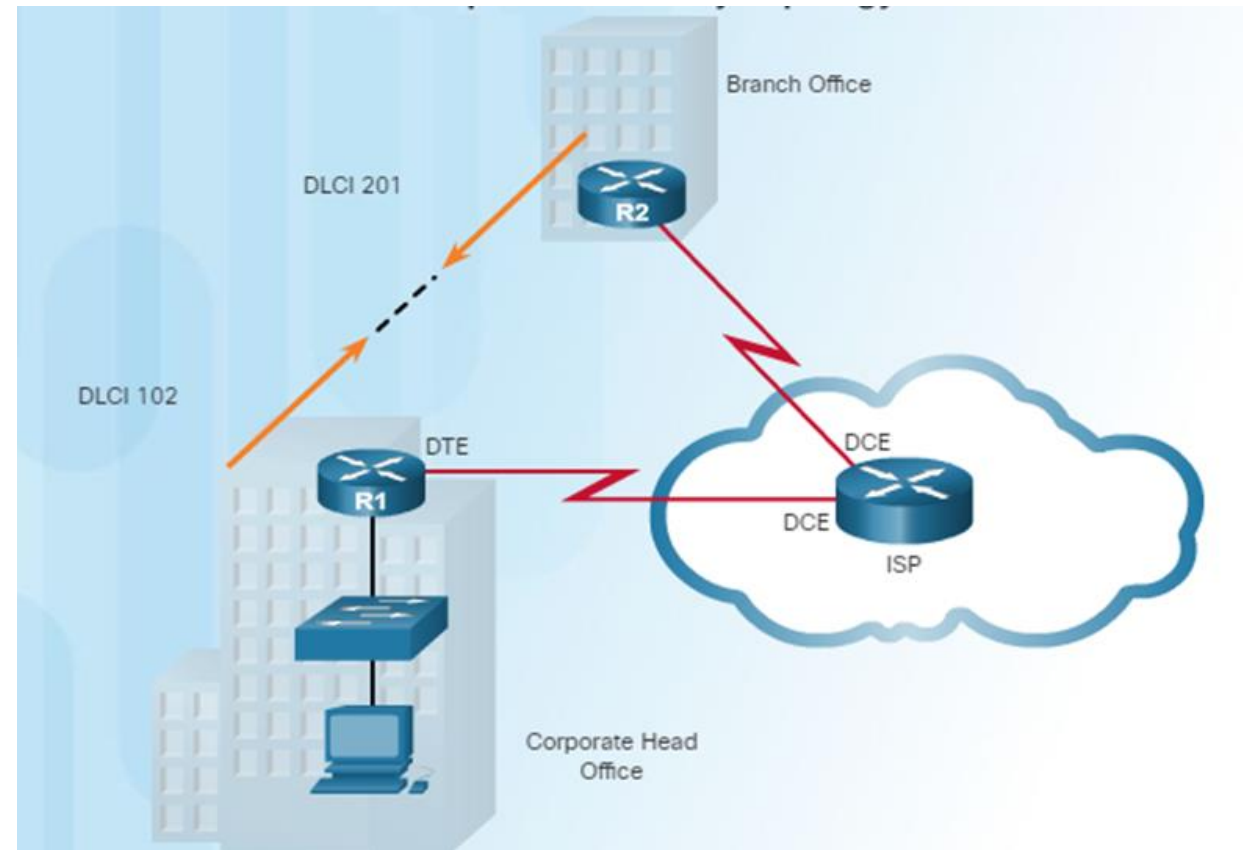
Private, Switched,
Packet switched,
Connection Oriented

- The Service Provider provides a **permanent virtual circuit (PVC)**, where the customer sees a continuous, dedicated connection
while the service provider figures out the route for each frame to its destination.
- Route is selected based on an identifier in the header.
- The identifier is called “a data-link connection identifier (DLCI)” which is unique for each PVC.

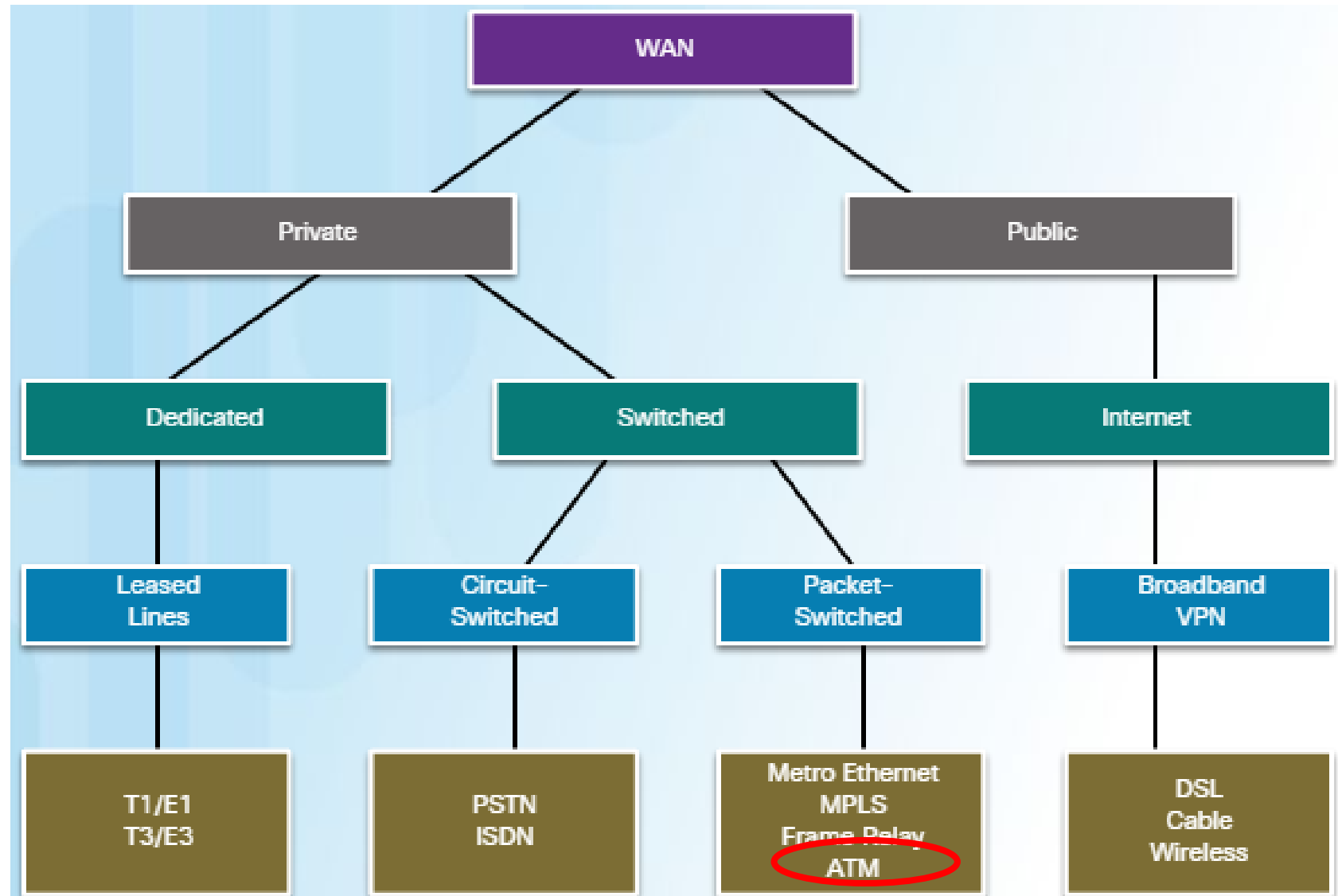
Private WAN Infrastructures cont.

Frame Relay Cont.

- A single router can be used to connect multiple sites using PVCs which carry both voice and data traffic.
- R1 will use DLCI 102 to reach R2
while R2 will use DLCI 201 to reach R1.



Private WAN Infrastructures cont.

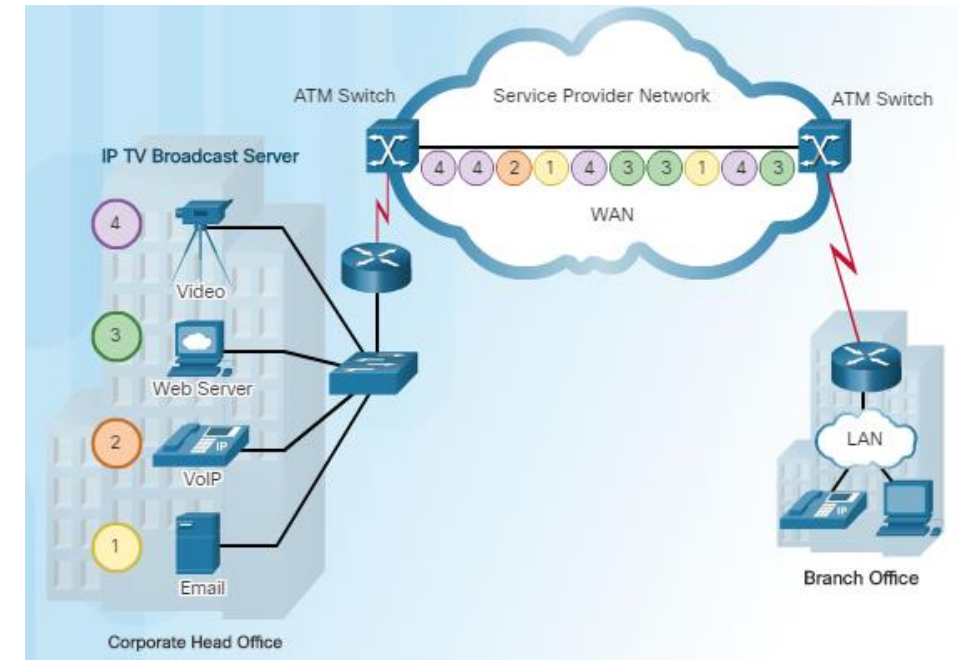


Private WAN Infrastructures cont.

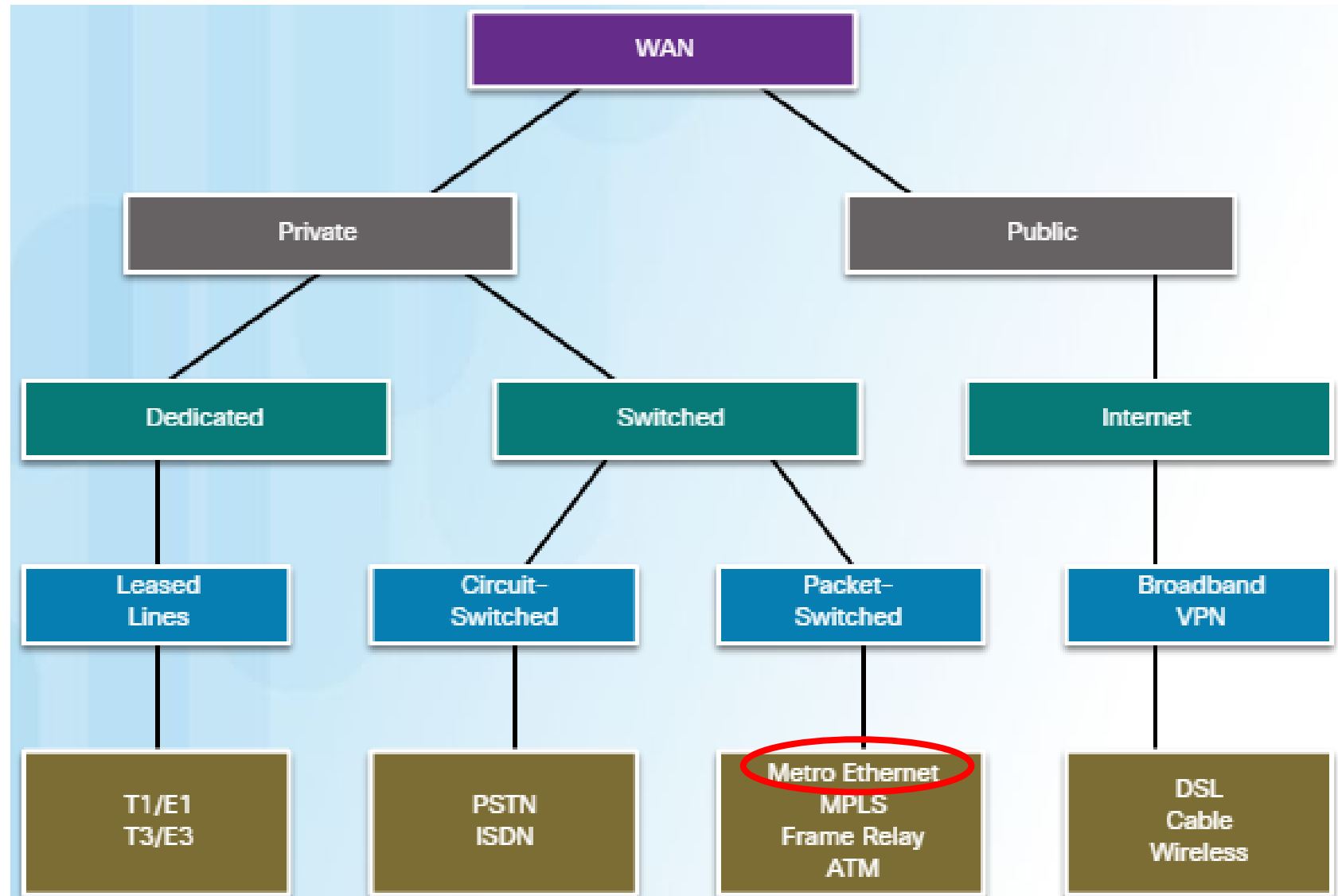
Asynchronous Transfer Mode (ATM)

Private, Switched,
Packet switched,
Connection Oriented

- Capable of transferring voice, video, and data through private networks.
- ATM uses asynchronous time-division multiplexing and encodes data into small, fixed-sized packets called cells (similar to ISO-OSI frames).



Private WAN Infrastructures cont.

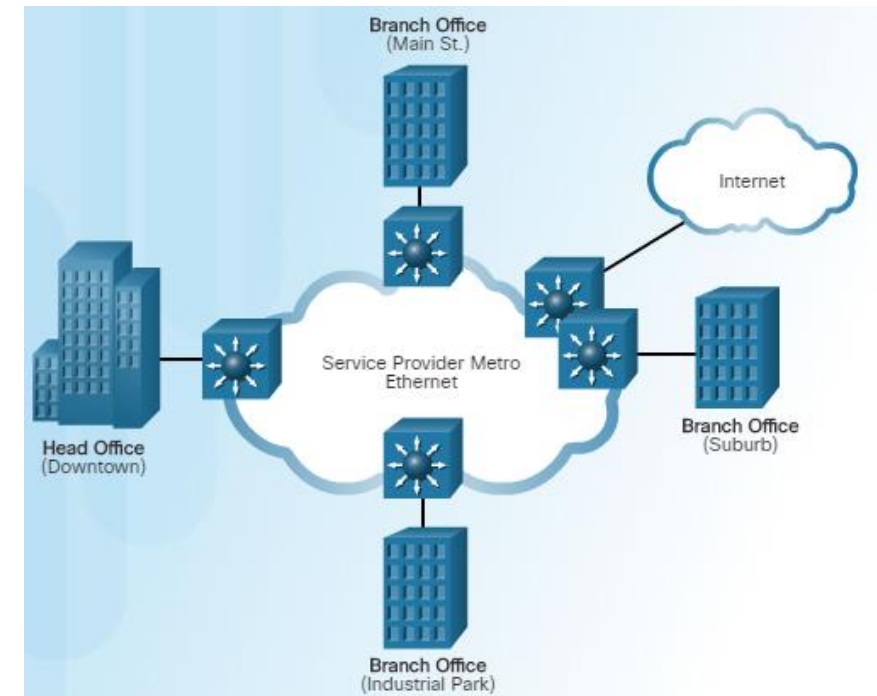


Private WAN Infrastructures cont.

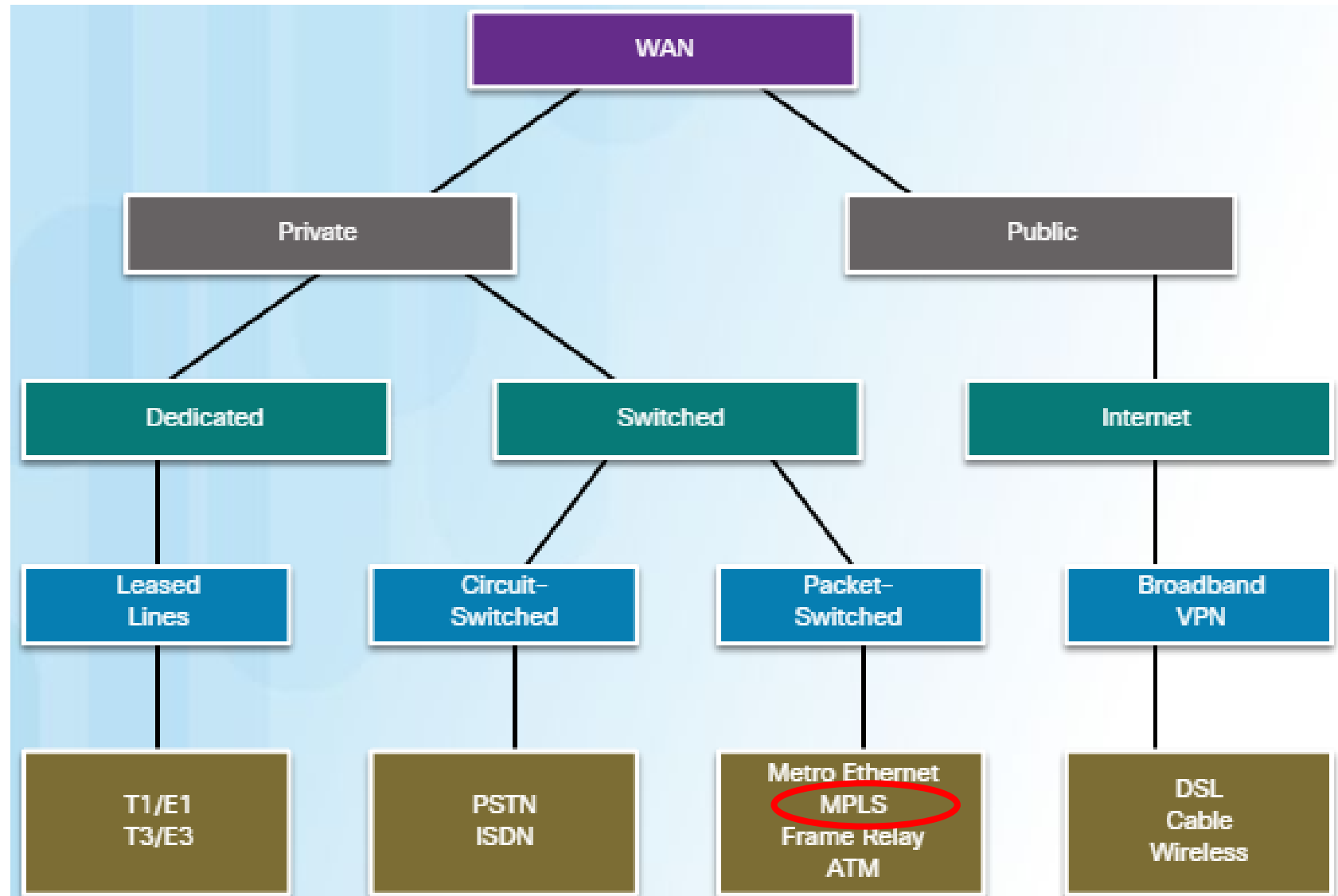
Ethernet WAN: Wide Area Ethernet (WAE)

- Service providers now offer Ethernet WAN service using fiber-optic cabling
- Known as fiber or LAN extension service.
- IEEE 1000BASE-LX standard supports fiber-optic cable lengths of 5 km
- IEEE 1000BASE-ZX standard supports cable lengths up to 70 km.
- Reduced expenses and administration
- Easy integration with existing networks
- Use to replace Frame Relay and ATM WAN links.

Private, Switched,
Packet switched,
Connection Oriented



Private WAN Infrastructures cont.

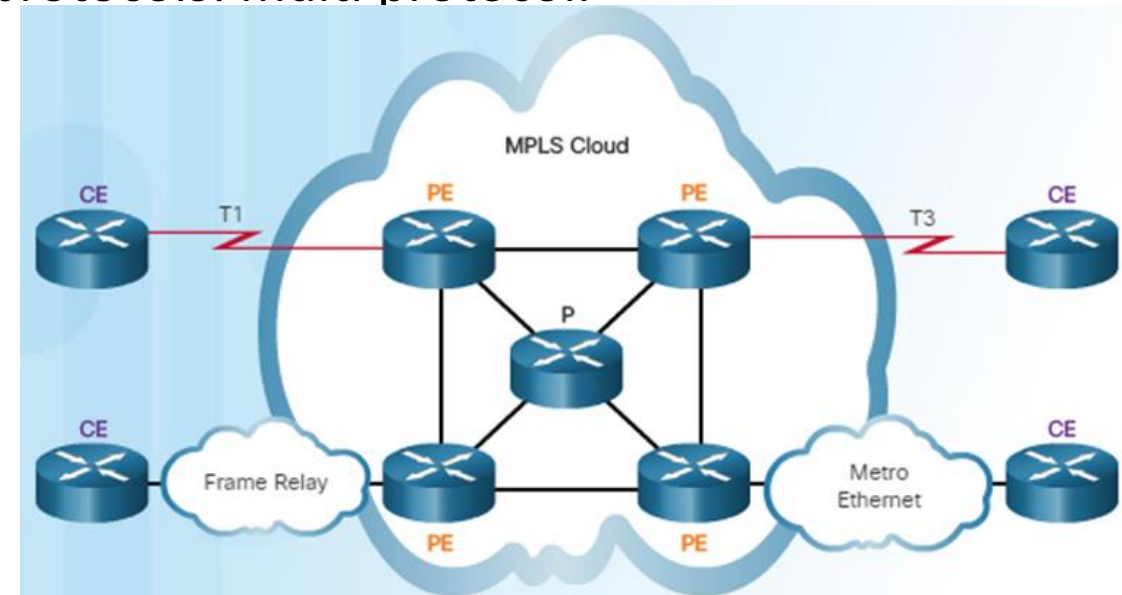


Private WAN Infrastructures cont.

Multiprotocol Label Switching (MPLS)

- MPLS is a service provider technology.
- Leased lines deliver bits, Frame Relay and Ethernet WAN deliver frames.
- However, MPLS can deliver any type of data unit.
- MPLS can encapsulate data units of various network protocols: multi protocol.

Private, Switched,
Packet switched,
Connection Oriented



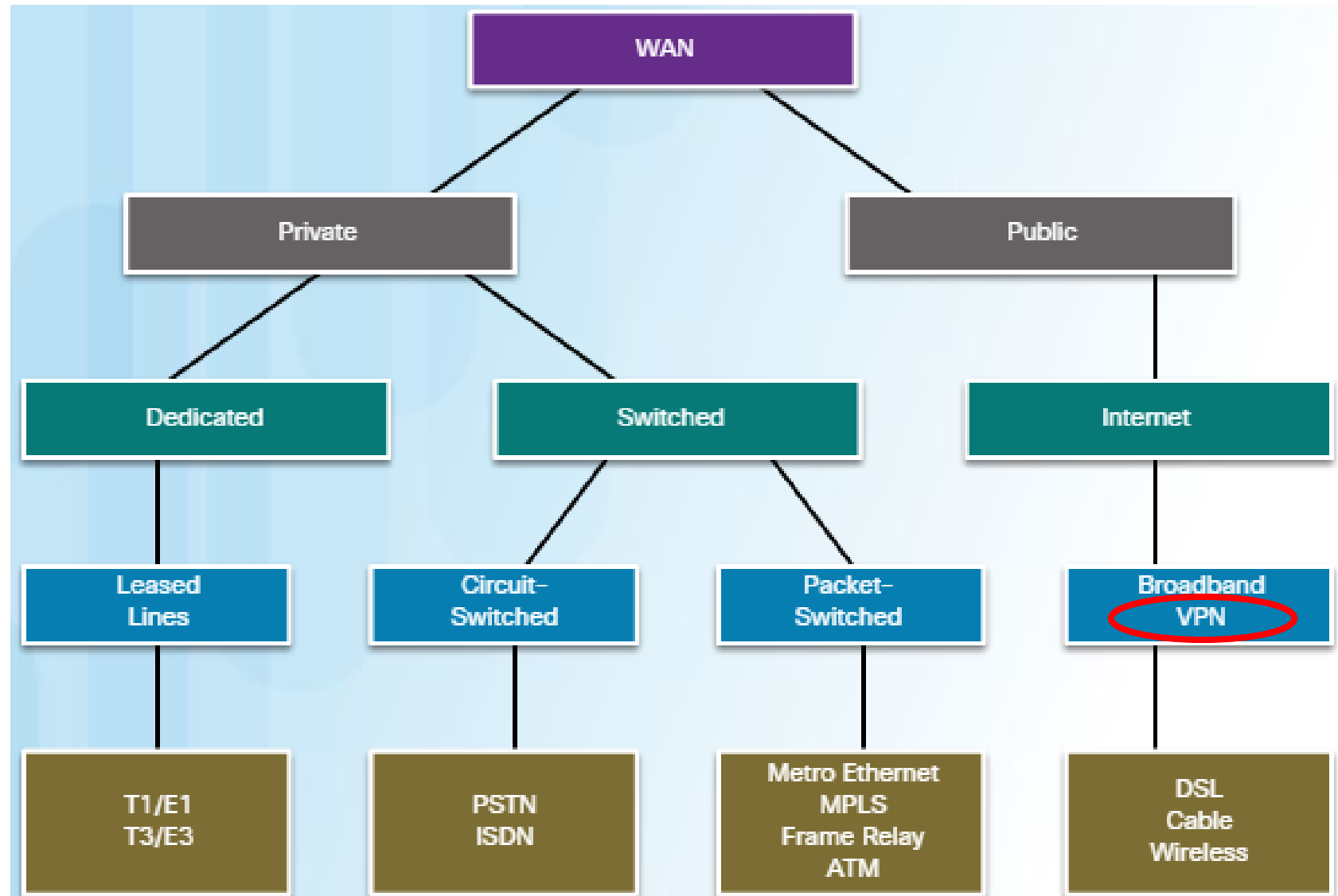
Private WAN Infrastructures cont.

Multiprotocol Label Switching (MPLS) Cont.

- In MPLS, traffic is categorized into forwarding equivalence classes (FEC) based on priority, QoS etc.
- The first time a packet enters the network, it's assigned to a specific FEC, indicated by appending a short bit sequence (the label) to the packet.
- Each router in the network has a table indicating how to handle packets (pre-determined path) of a specific FEC type.
- So subsequent routers use the label as an index, and use the pre-determined path given in the table to forward the packet.

Private, Switched,
Packet switched,
Connection Oriented

public WAN Infrastructures

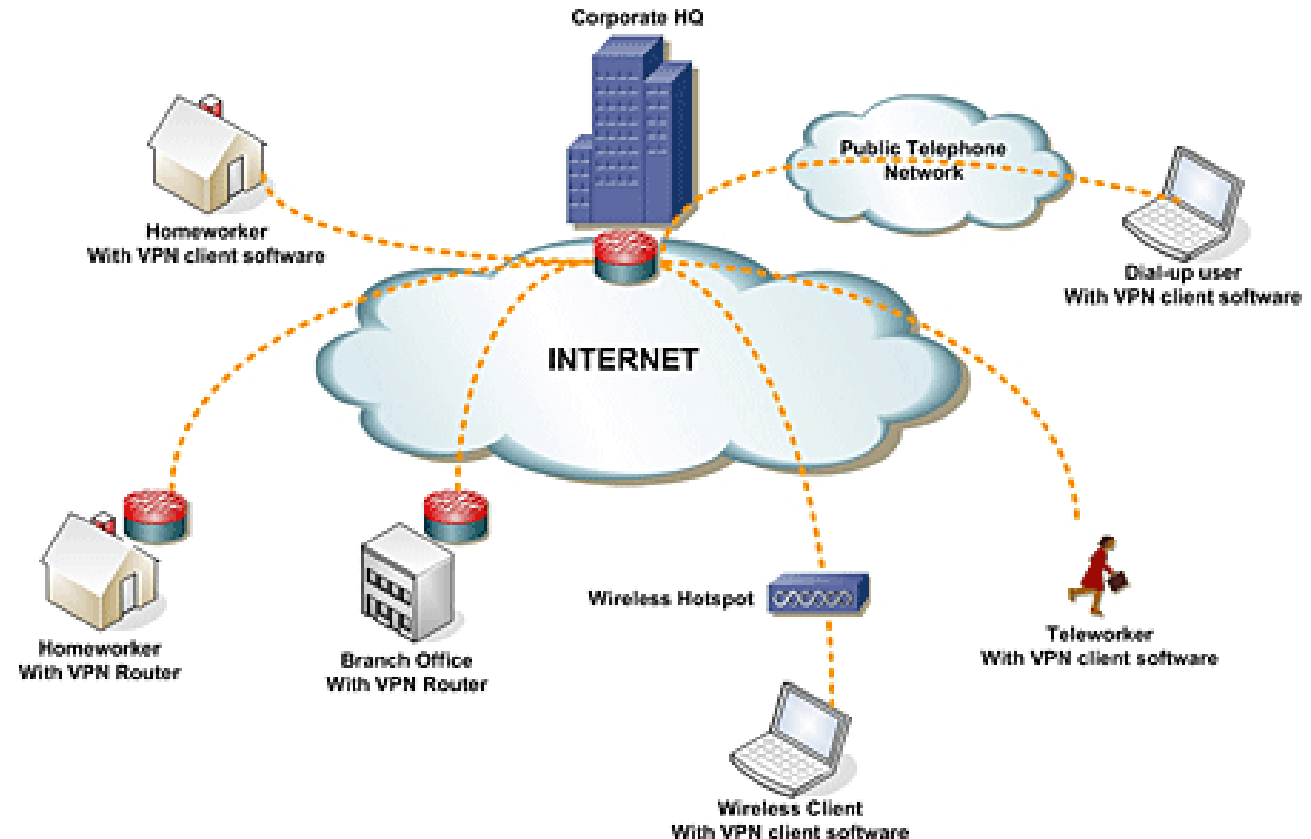


public WAN Infrastructures cont.

Public, Internet
VPN

Virtual Private Networks (VPNs)

- An encrypted connection between private networks over a public network, such as the Internet.
- Compatibility with broadband technology
- Cost savings
- Security
- Scalability

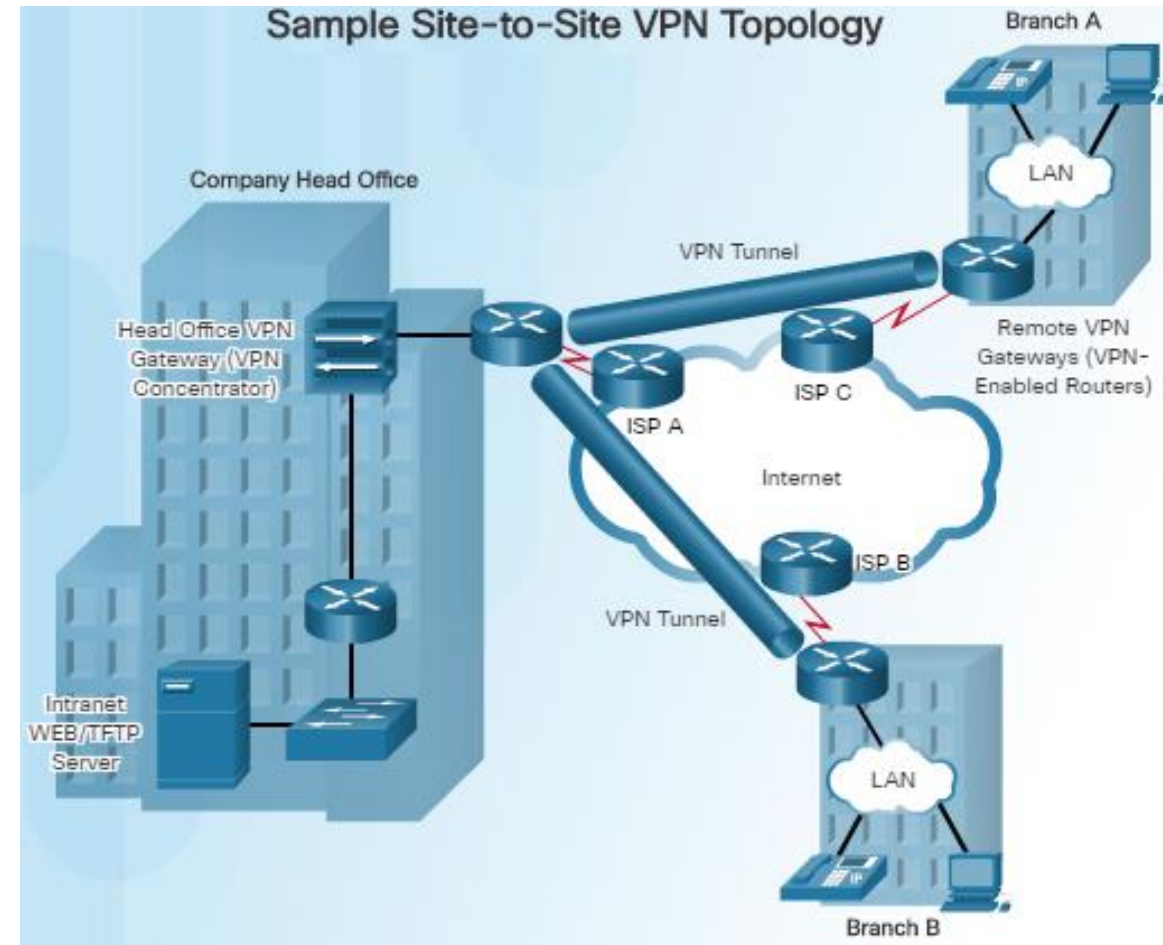


public WAN Infrastructures cont.

Virtual Private Networks (VPNs)

- Site-to-Site VPN
 - ✓ Connects entire networks to each other
 - ✓ Ex: connect a branch office network to a company headquarters network

Public, Internet
VPN

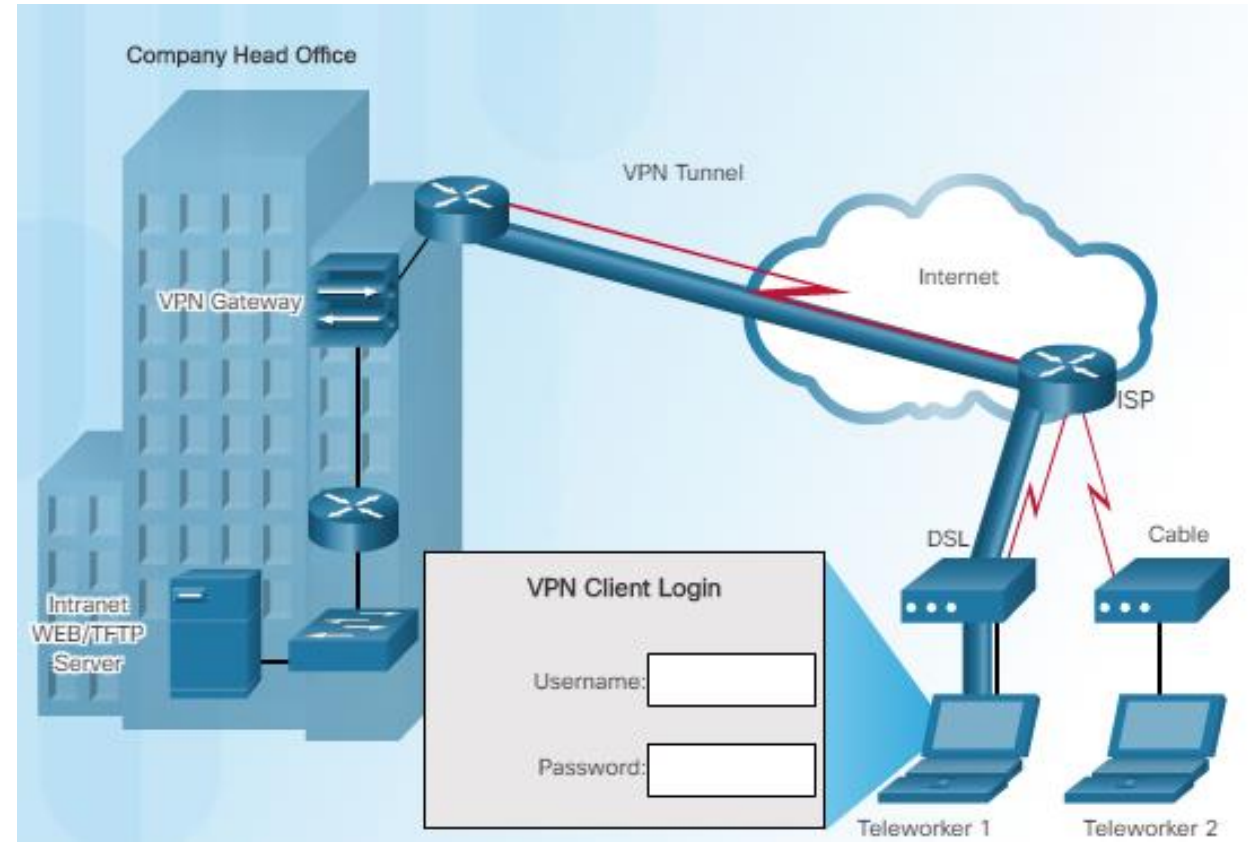


public WAN Infrastructures cont.

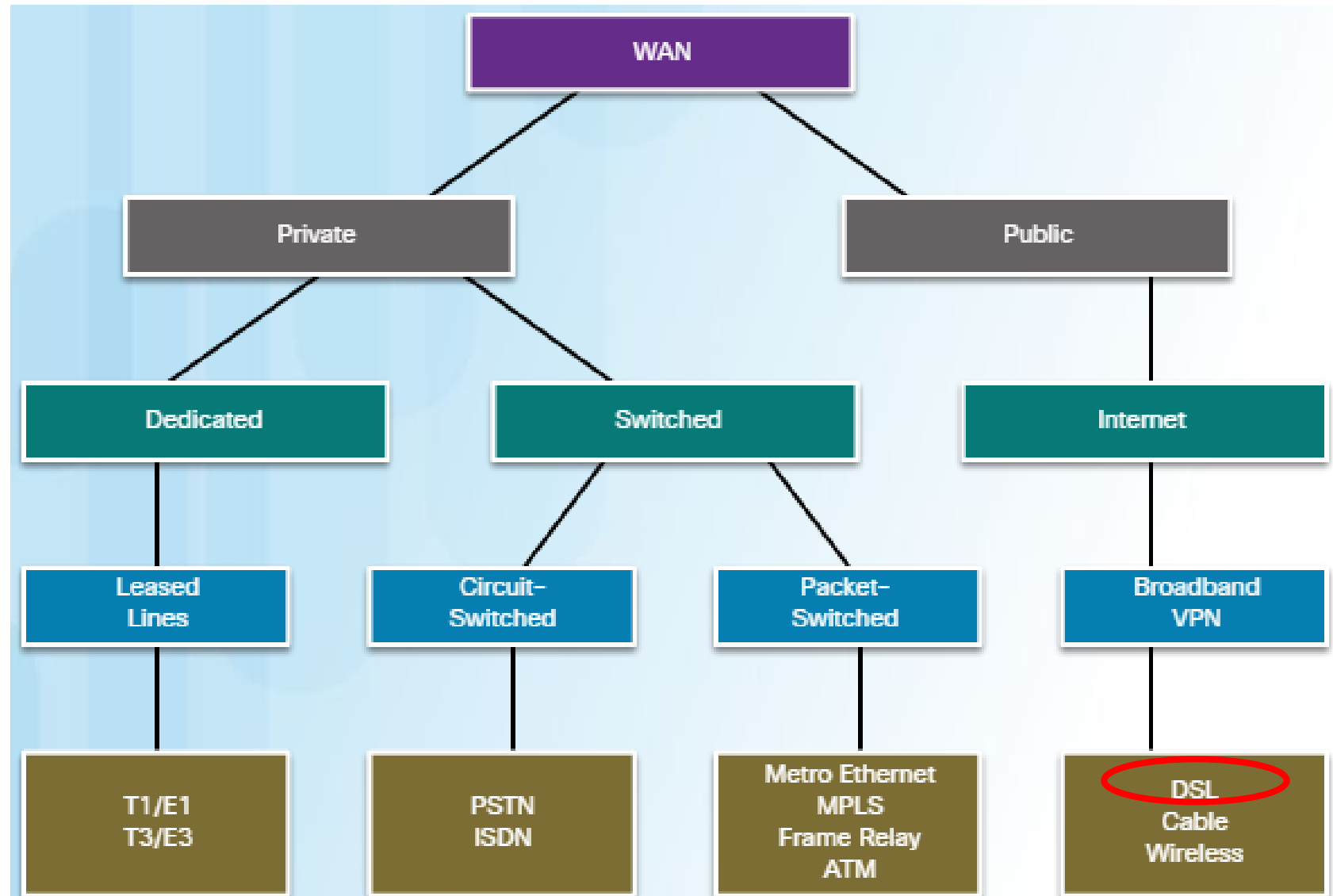
Public, Internet
VPN

Virtual Private Networks (VPNs) Cont.

- Remote VPN
 - ✓ Enables telecommuters, mobile users, and extranet consumers to access a company network securely over the Internet.



public WAN Infrastructures cont.



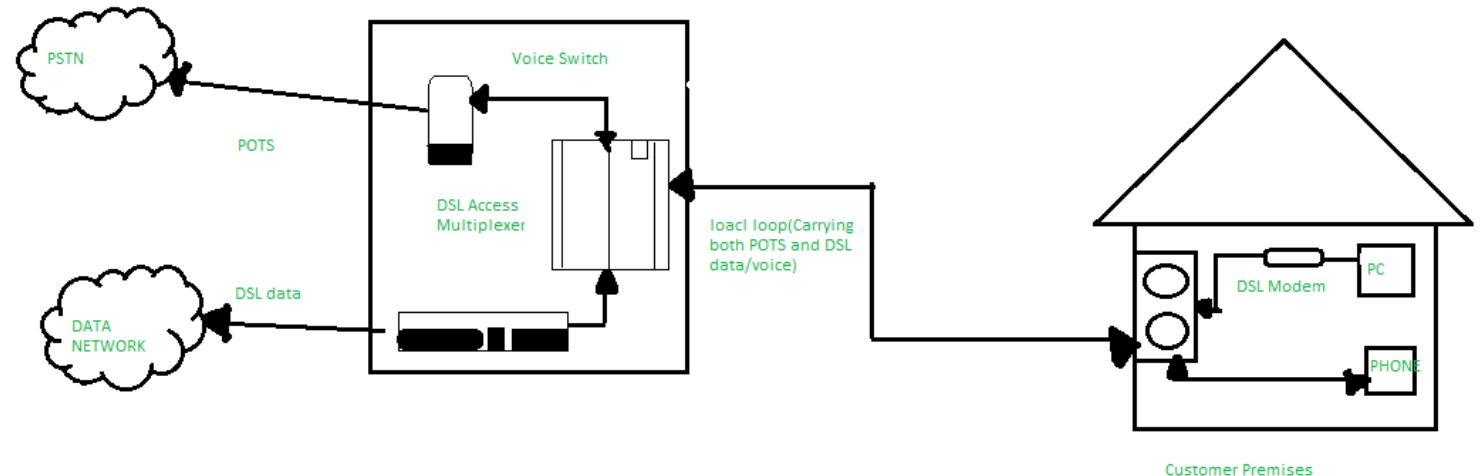
public WAN Infrastructures cont.

Public, Internet
Broadband

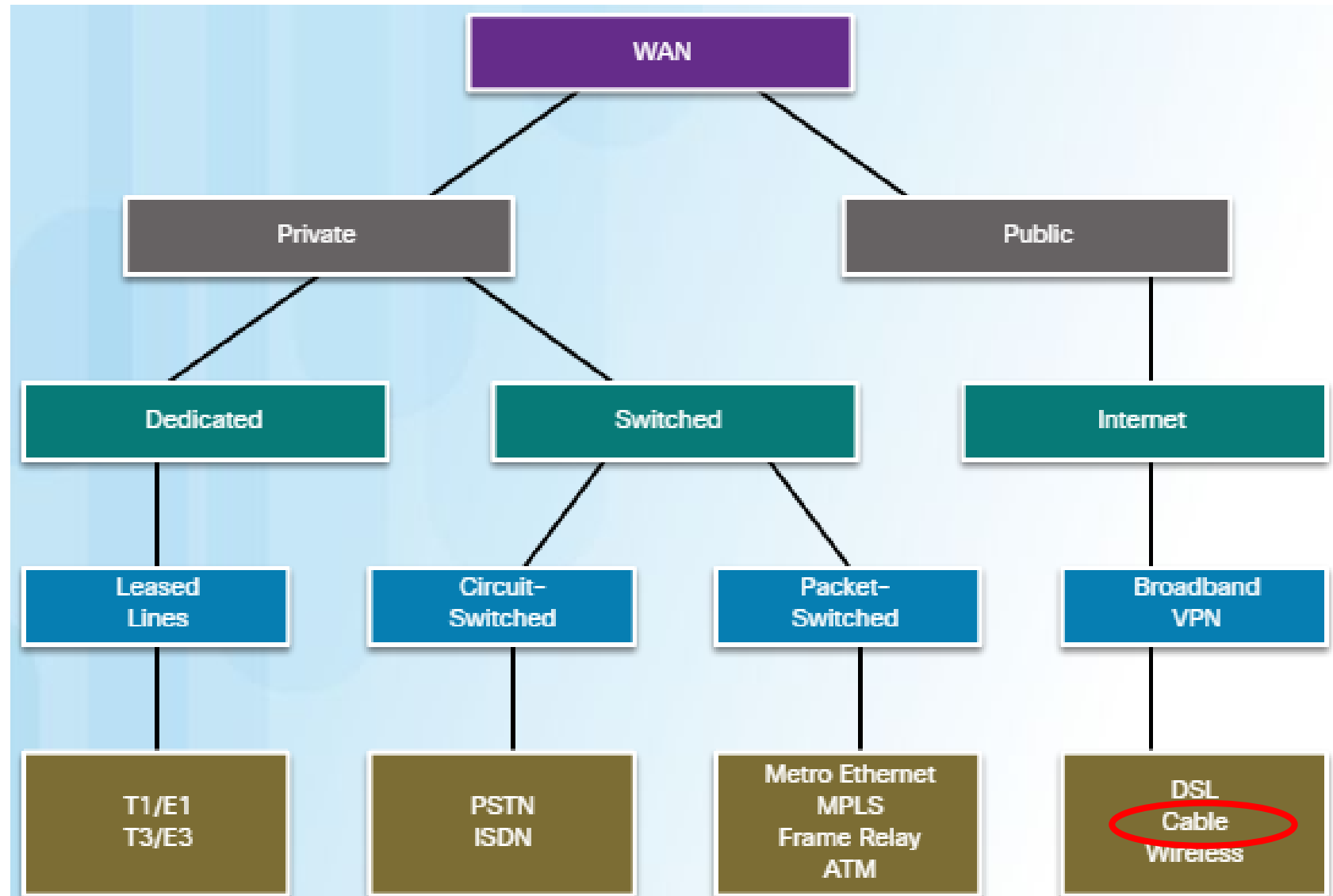
Digital Subscriber Line (DSL)

- Digital transmission of data over the existing PSTN.
- Voice signal is transmitted using low frequencies from 0 Hz to 4 kHz: voice band.
- Digital signals (data) is transmitted using high frequencies from 25 kHz to 1.5 MHz.
- DSL filters or splitters are used divide the signals.

- DSL is around 250 times faster
than ISDN
- DSL is always on
ISDN is dial-up



public WAN Infrastructures cont.

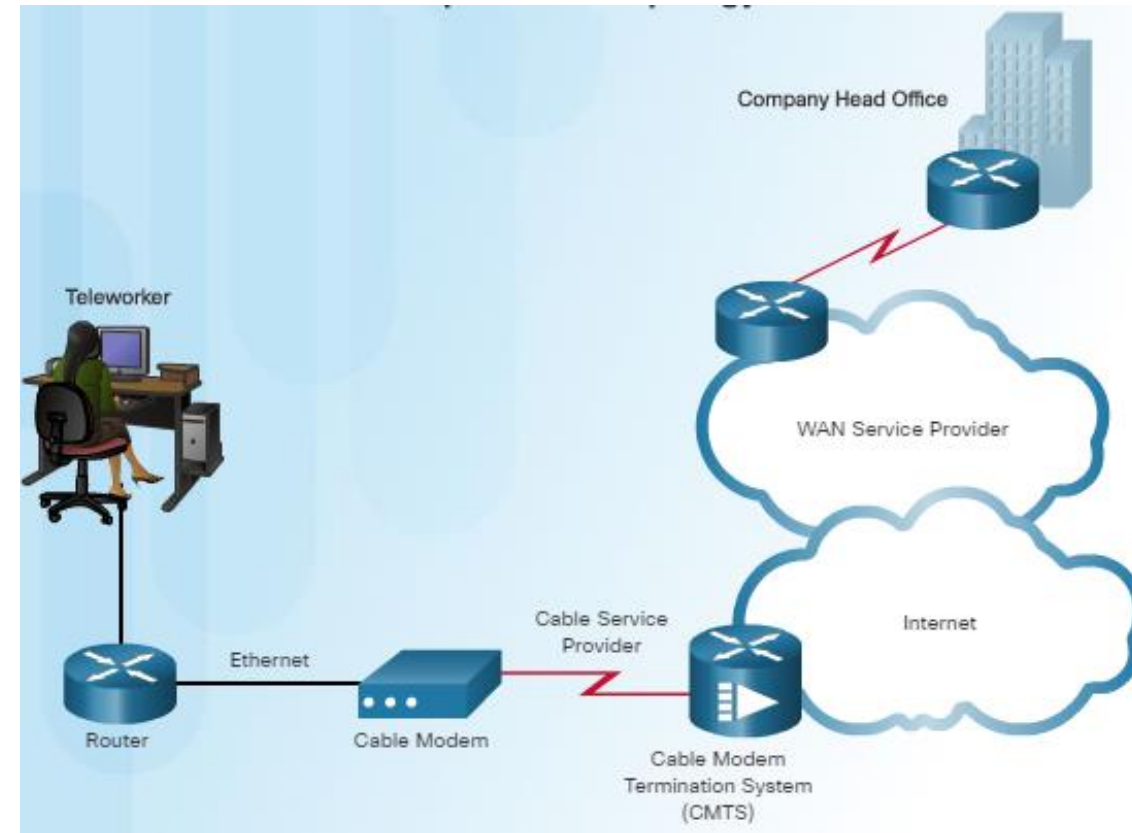


public WAN Infrastructures cont.

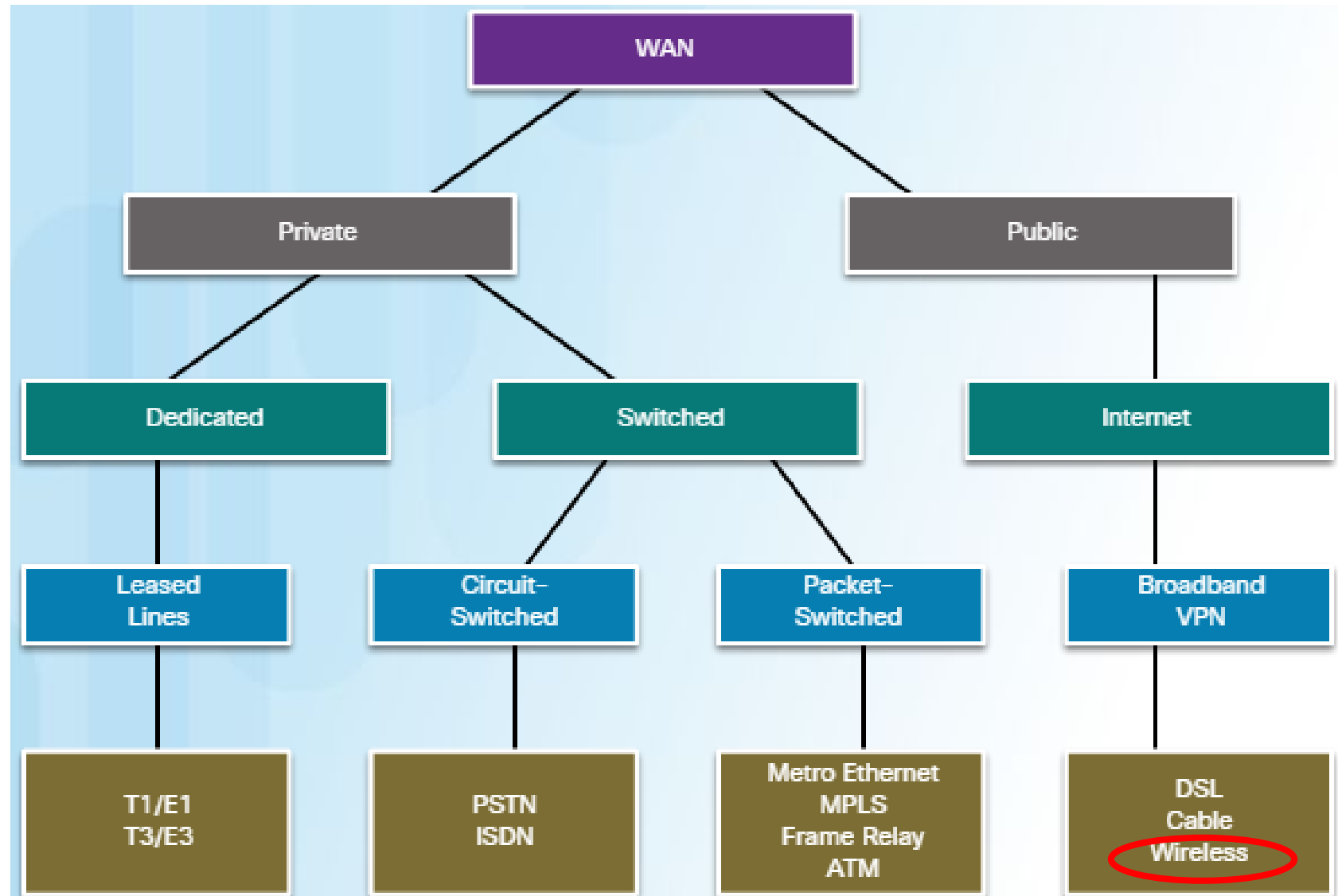
Cable

- Network access is available from many cable television providers.
- A subscriber connects a computer / LAN router to the cable modem, which translates the digital signals into broadband frequencies used for transmitting on a cable television network.

Public, Internet
Broadband



public WAN Infrastructures cont.

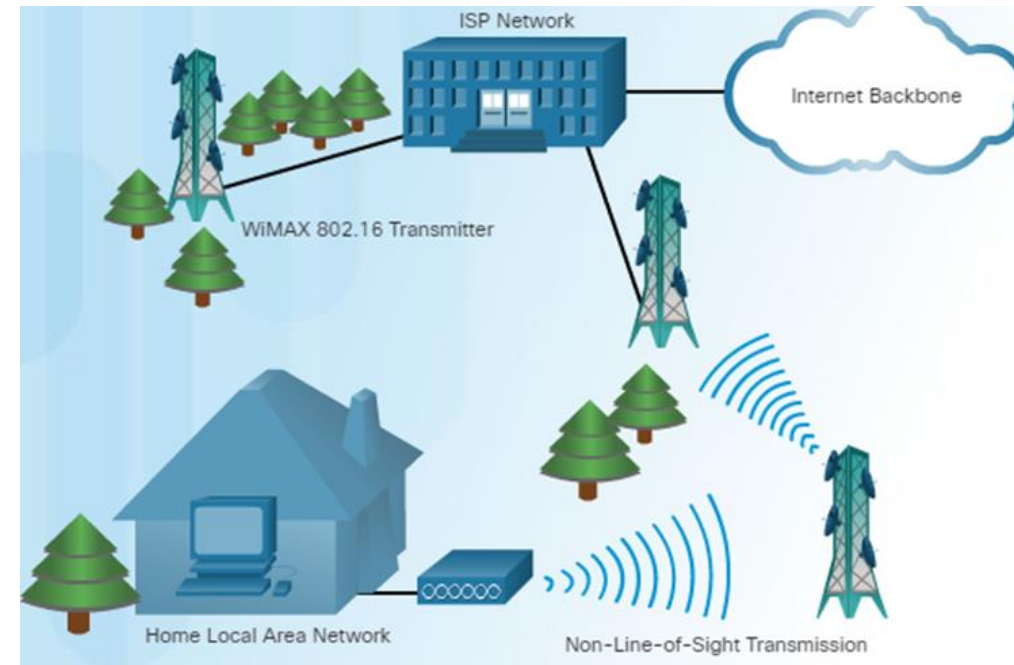


public WAN Infrastructures cont.

Public, Internet
Broadband

Wireless

- Uses the unlicensed radio spectrum to send and receive data
- WiMAX - Worldwide Interoperability for Microwave Access
- Municipal Wi-Fi: municipal wireless networks
- Very small aperture terminal (VSAT): Satellite Connections

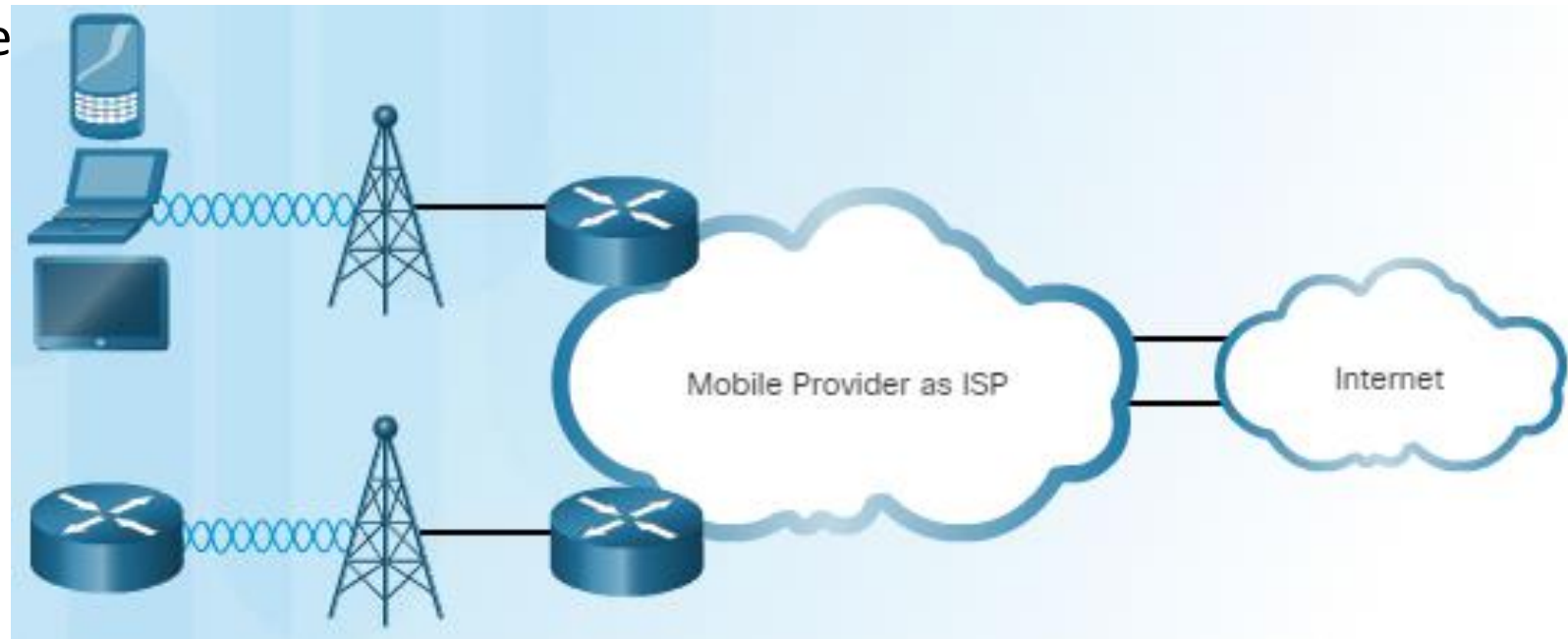


public WAN Infrastructures cont.

Public, Internet
Broadband

3G/4G Cellular

- Phones, tablet computers, laptops etc. communicate through the Internet using cellular technology.
- Use radio waves to communicate through a nearby mobile phone tower.
- Long-Term Evolution (LTE) – Fast



Choosing a wan link connection

- What is the purpose of the WAN?
- What is the geographic scope?
- What are the traffic requirements?
- Should the WAN use private or public infrastructure?
- For a private WAN, should it be dedicated or switched?
- For a public WAN, what type of VPN access is required?
- Which connection options are available locally?
- What are the costs of the connection options available?